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TECHNOLOGIES, INVESTMENT STRATEGIES IN
THE FUTURE REVOLUTION

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2023

BLOCKCHAIN 2023 INVESTING

WEB 3.0 COINS, A.I., CRYPTO, BITCOIN, BLOCKCHAIN-
AS-A-SERVICE (BAAS), BUY STOCKS IN BLOCKCHAINS
DISRUPTIVE TECHNOLOGIES, INVESTMENT
STRATEGIES IN THE FUTURE REVOLUTION

NFT TRENDING CRYPTO ART

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INTRODUCTION

Welcome to the world of Blockchain! In this book, we will explore the exciting world of Blockchain technology and its potential to revolutionize how we interact and do business. You may have heard of Blockchain and cryptocurrencies like Bitcoin or Ethereum, but the technology can do much more than you think.

From enabling secure and transparent record-keeping to facilitating peer-to-peer transactions without need for intermediaries, Blockchain has the potential to disrupt and modify a wide range of industries. In this book, we will delve into the history and principles of Blockchain, as well as its current and potential future applications.

Whether you are a complete newcomer or an experienced professional looking to deepen your understanding of this innovative technology, this book has something to offer everyone. We will cover the technical aspects of Blockchain in an accessible and easy-to-understand manner, making it suitable for readers at all levels of expertise.

Blockchain technology is transforming how we do business, manage our finances, and interact with each other. With its revolutionary distributed ledger technology, Blockchain offers unprecedented transparency, security, and trust in transactions.

This book will provide an in-depth look at the technology, its applications, and the many ways it is already revolutionizing the world. We explore the fundamentals, use cases, and the opportunities it presents for businesses, governments, and individuals. You'll also learn about the challenges and risks associated with the technology and the steps to ensure

successful adoption and use.

We'll dive deep into the world of cryptocurrency, and the potential of Blockchain to create new markets, monetize data, and make financial transactions more efficient and secure. We'll also discuss its potential to develop new governance models and disrupt the traditional business model.

by the end of the book, you'll have a better understanding of the technology and its potential, including the necessary tools to begin building your own applications and projects.

So buckle up. Join us on this journey to learn the exciting world of Blockchain and its boundless potential to shape the future.

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WHAT IS BLOCKCHAIN TECHNOLOGY?

HOW DOES BLOCKCHAIN WORK?

The term "Blockchain technology" has probably come up with cryptocurrencies like Bitcoin over the past few years. Maybe you're thinking, "What is Blockchain technology exactly?" Blockchain is a platitude, but only in the hypothetical sense because it lacks a tangible meaning, one that the typical person can clearly understand. It is essential to explain "what Blockchain technology is," including the technology involved, how it functions and is becoming important in the digital sphere.

You are responsible for educating yourself on this developing technology to be ready for the future when Blockchain develops and becomes more approachable. If you're new to Blockchain, this is the best location to learn the basics. In this chapter, you will discover how to respond to the query, "what is Blockchain technology?" Additionally, you will find out how Blockchain functions, why it's significant, and how you may use it to advance your profession.

What is Blockchain?

To comprehend Blockchain, you must know that a block is only a piece of data. Anything that may be stored digitally, like financial transactions, invoices, bills of lading, or hashes from the block previously added to the chain is included in that block.

What is a hash? Consider it a statistically determined fingerprint that identifies every piece of information in that block. The fingerprint of the previous block is included in every block, and this chaining of hashes gives the Blockchain its immutable characteristics. The hash of that block and all succeeding hashes will change if someone went back and changed a transaction in the chain.

What is Blockchain technology?

Blockchain technology stores data that makes it difficult or impossible for the system to be changed, hacked, or misused. A distributed ledger known as a Blockchain distributes and copies transactions among a system of computers.

Blockchain technology keeps public transactional records across numerous databases in a network connected by peer-to-peer nodes. These records are commonly referred to as "blocks," and the term "digital ledger" is widely used to describe this storage.

The owner's digital signature serves as a validation and fraud prevention mechanism for every transaction in this ledger and helps authenticate the transaction. The information in the digital ledger is, therefore, very secure.

The digital ledger can be compared to a network of computers sharing a Google spreadsheet, whereas transactional data are preserved per actual purchases. The data is available for public viewing, but it cannot be changed, which is a fascinating feature.

How Blockchain Started

The term "Blockchain" may conjure images of Bitcoin or other cryptocurrencies. This was the setting where the term was first introduced; but before all of that, there was the idea of a Blockchain.

David Chaum suggested a Blockchain-like system in his doctoral paper at the University of California, Berkeley, back in 1982.

Bellcore researchers submitted a report on using a Blockchain technique to ensure document timestamps nearly ten years later. They would later start a business called Surety (more on them in a second). Before 2008, a person or group, Satoshi Nakamoto, developed the Blockchain idea that would eventually power Bitcoin.

Relating to Surety, since 1995, it has been posting the most recent hash of each electronic document it has ever sealed in The New York Times Public Notices section. The fact that it is generally known taps into another approach that Blockchains can use. It would be challenging to covertly edit a block that has already been added to the chain, the more widely broadcast a hash is.

Since each participant has a copy of the most recent hash, this idea maintains integrity in public and consortium Blockchains. It ultimately evolved into Bitcoin's core, the well-known cryptocurrency that acts as a public record for all network transactions.

Why is Blockchain Popular?

Consider sending funds from your bank account to relatives or friends. You connect to online banking using your account number and transfer the funds to another individual. Your bank updates the transaction records after the transaction is complete. It is relatively easy. But a potential problem exists that most of us ignore: these kinds of transactions are easily manipulated. People aware of this truth are reluctant to use them, which is the reason why third-party payment programs have developed over the past few years. This weakness was the primary motivation behind the development of Blockchain technology.

In terms of technology, a Blockchain is a digital ledger that has recently attracted much interest. But why has it gained such a following? So let's investigate it to understand the idea entirely. Data and transaction recording is an essential component of the business. When this information is handled internally or through third parties like brokers, bankers, or lawyers, the business takes time, money, or both. Fortunately, Blockchain eliminates this drawn-out procedure and enables the transaction to move more quickly, saving time and money.

Most people believe Bitcoin and Blockchain may be used interchangeably, but it is different. Bitcoin is a currency that depends on Blockchain technology to be secure, and Blockchain technology can enable applications connected to multiple industries like finance, supply chain, manufacturing, etc.

Blockchain is a new technology with several benefits in a society that is becoming more digital:

- **Extremely secure:** It uses a digital signature function to execute fraud-free transactions, making it easier for other users with a specific digital signature to corrupt or edit an individual's data.
- **Decentralized system:** In the past, the approval of regulatory bodies like a government or bank is necessary for transactions. Still, with Blockchain, transactions are completed by user consensus, resulting

in trouble-free, safer, and faster transactions.

- **Automation capability:** When the trigger's requirements are satisfied, it can be programmed to automatically generate a series of activities, events, and payments.

How does Blockchain technology work?

You may have observed that several companies have been incorporating Blockchain technology in recent years. But how does this technology operate? Is it a substantial modification or merely an addition? Let's start with demystifying Blockchain technology since it is still in its early stages and has the potential to be revolutionary in the future.

Combining three popular technologies, Blockchain consists of:

1. Cryptographic keys
2. A peer-to-peer network with a shared ledger
3. A method of computing that can be used to store network transactions and records

Two keys make up a cryptography key: private and public keys. These secrets aid in the execution of successful transactions involving two parties. These two keys are unique to each person and used to create a secure digital identity reference. The most significant component of Blockchain technology is this protected identification. This identity is known as a "digital signature" and is used to approve and manage transactions.

The peer-to-peer network and the digital signature are combined, and many people who serve as authorities use the digital signature to agree on transactions and other matters. As soon as they approve a transaction, it is mathematically verified to ensure it is valid, which leads to a successful secured transaction between the two network-connected parties. In conclusion, cryptography keys are used by Blockchain users to conduct various kinds of digital exchanges across the peer-to-peer network.

Different Types of Blockchain

Blockchains come in four different varieties:

Private Blockchain networks

A single authority controls a private Blockchain network that functions for private corporations and organizations on closed networks. Private Blockchains allow businesses to set network characteristics, accessibility and permission choices, and other crucial security features.

Public Blockchain networks

Instead of being kept in one place, data is spread throughout a peer-to-peer network using DLT. A consensus algorithm verifies the legitimacy of information; the two popular consensus techniques are Proof of Stake (PoS) and Proof of Work (PoW). Public Blockchains helped spread awareness of distributed ledger technology (DLT) and removed some difficulties and problems, including centralization and security weaknesses.

Permissioned Blockchain networks

Permissioned Blockchain networks, sometimes called hybrid Blockchains, are private Blockchains that grant approved users exclusive access. These Blockchains are frequently set up by businesses to achieve the best of both worlds. They enable better structure when determining who can participate in the network and what transactions.

Consortium Blockchains

Consortium Blockchains are also the best for working with various organizations. Similar to permissioned Blockchains, consortium Blockchains feature both public and private components. However, a single consortium Blockchain network will be managed by numerous companies. Though initially more difficult to set up, these Blockchains can provide superior security once operational.

The Process of Transaction

The manner that Blockchain technology confirms and authorizes transactions is one of its defining features. For example, the first party would attach the transaction details to the second party's public key if they were to complete a transaction using their respective private and public keys. Then, this entire data is collated into a block.

The block comprises a timestamp, a digital signature, and other vital information. It should be emphasized that the block does not contain information on both parties' transaction identities. The block is then sent around the entire network, and the transaction is completed when the right user uses his private key to match it with the block.

An example of how Blockchain is put to use:

Hash encryptions

Blockchain technology hashes and encrypts data, mainly using the SHA256 method to protect the data. The SHA256 method conveys information about the sender's address (public key), the recipient's address, the transaction, and the private key. After being verified, the hashed data is sent around the globe and put into the Blockchain in encrypted form. The sender and receiver's authentication is made more accessible by the SHA256 algorithm's virtually impenetrable hash encryption.

Proof of work

- Each block in a Blockchain has four major headers.
- Previous hash: addresses identifies the block before this one.
- Transaction details: information about each transaction that must take place.
- Nonce: in cryptography, an arbitrary integer is used to distinguish the block's hash address.
- Block hash address: a hashing algorithm is used to send all the information mentioned above, including the transaction information and nonce. This produces an output that is the distinct "hash address," which is a 256-bit number with 64 characters in length. As

a result, it is referred to as the block hash.

- Using computer procedures, many people attempt to discover the appropriate hash value to satisfy predetermined criteria. When the predetermined condition is met, the transaction is finished. To put it simply, Blockchain miners attempt to solve a proof of work problem, a mathematical puzzle, and the first person to solve it earns a prize.

Mining

Adding transactional information to the current digital/public ledger is known as "mining" in Blockchain technology. Despite being linked to Bitcoin, the phrase is also used to describe other Blockchain technologies. Mining involves creating a block transaction hash that is difficult to forge. This ensures the security of the entire Blockchain without needing a centralized system.

Blockchain: Advantages and Disadvantages

Blockchain technology has several benefits and drawbacks:

Advantages

- As a result of their high level of security, one of the key benefits of Blockchains is that they can secure and protect sensitive data from online transactions.
- Blockchain technology also provides quick and convenient transactions for those who desire them.
- In contrast to other transaction techniques, which can take several days to complete, it only takes a few minutes.
- There is no third-party involvement from governmental or financial entities, which many users view as a positive.

Disadvantages

- Blockchain and cryptography both use public and private keys, and private keys are said to have occasionally caused problems.
- One drawback of Blockchains is that users need help if they misplace their private keys.
- Another is the scaling limitations resulting from the cap on transactions per node, and because of this, finishing numerous transactions and other tasks can take several hours.
- The difficulty of adding or changing information once it has been recorded is another significant flaw with Blockchain technology.

How is Blockchain used?

Blockchains hold data on cryptocurrency-based financial transactions and other kinds, like product tracking. For instance, food products can be tracked from when they are shipped out until they are delivered to their destination. This information can be helpful since it makes it simple to identify the source of a contamination epidemic. This is just one of the numerous ways Blockchains may be used to hold crucial data for businesses.

Hyperledger, hosted by the Linux Foundation

The Linux Foundation is the home of the international partnership known as Hyperledger, which brings together leaders in technology, supply chain, banking, and IoT (Internet of Things). By establishing an open, cross-industry standard for distributed ledgers, Hyperledger Fabric enables programmers to create Blockchain applications that cater to particular requirements.

Decentralization

Decentralization is the distribution or dispersion of functions, powers, people, or things far from a central place or authority. In business, it often refers to the transfer of power from top executives to middle managers and lower team members on the organizational chart. Devolution has many advantages, but the ones mentioned most frequently include better communication, enhanced employee empowerment, and improved flexibility and responsiveness.

Transparency

Transparency is one of the most critical parts of decentralization. All employees have permission to access information and decision-making procedures in a decentralized organization. Because of this openness, workers are more inclined to trust one another and work together. Additionally, it

enables workers to hold bosses responsible for their choices.

Bitcoin vs. Blockchain

The most well-known and profitable cryptocurrency to date, Bitcoin, was first presented in 2009 and has since grown in popularity. Due to its decentralized nature, which prevents a central authority or bank from managing its supply, Bitcoin is extremely popular. This also means that using it entails no transaction fees and anonymous transactions.

Blockchain is a database of exchanges between two parties, with blocks of data, including details about each exchange being added to the chain in the order in which they occur. The Blockchain constantly expands as new blocks are added to it, and as more blocks are added after each record, it becomes harder to edit that record.

Blockchain vs. Banks

The financial sector could transform because of Blockchain. Blockchain allows banks to keep up with the shifting demands of the digital era by enabling them to react more quickly. Banks may give their clients a safer and more effective way to conduct transactions using Blockchain. Blockchain can also assist banks in streamlining their processes and cutting costs.

Blockchain: Why is it Important?

Blockchain is important because it can change the financial industry. In fact, it allows banks to keep up with the shifting demands of the digital era by enabling them to react more quickly. Banks may give their clients a safer and more effective way to conduct transactions using Blockchain. Blockchain can also assist banks in streamlining their processes and cutting costs.

What is a Blockchain platform?

A Blockchain platform is a distributed digital ledger that enables users to store and exchange information securely. A distributed network of computers retains the register, where each transaction is confirmed by agreement among network users.

Proof of Work (PoW) vs. Proof of Stake (PoS)

Blockchain is a continuously growing distributed database that maintains track of a set of records called blocks. It's frequently claimed that Blockchain technology has the power to transform a wide range of sectors, including finance, law, and healthcare. The Proof of Work (PoW) algorithm is used to build blocks and safeguard the Blockchain. In return, miners must solve a riddle to make a block and get the block reward.

An alternative technique that secures the Blockchain without the need for mining is called Proof of Stake (PoS). Instead, to be eligible for rewards, users must lock up a portion of their money for a set period.

What Advantages do Blockchains Have Over Traditional Finance?

Compared to traditional finance, Blockchain has several potential advantages. The fact that Blockchain is decentralized, whereas traditional finance is centralized, is one of its most lauded benefits. This implies that a Blockchain system has no single point of failure. Blockchain also has the benefit of being more transparent than conventional finance.

Investment strategies for Blockchain technology

The opportunity to invest in Blockchain technology is always available. There are numerous methods to move forward with buying your first Blockchain investment, and stocks and Blockchain technology can be profitable investments. When investing in Blockchain technology, Bitcoin is frequently the first thing that springs to mind and shouldn't be disregarded.

In addition to Bitcoin, there is the opportunity to invest in cryptocurrency penny stocks like Litecoin and Altcoin. Additionally, several pre-development apps and services are raising money through Blockchain technology. As an investor, you might purchase coins with the hope that they will appreciate it if the service or application gains popularity. Moreover, buying stock in startups using Blockchain technology is another option to invest in the technology.

Investment methodologies using Blockchain and traditional finance

The two basic investment approaches in conventional finance are active and passive. Selecting stocks or other assets and hanging onto them for a long time constitute active investment. Contrarily, passive investing entails purchasing various assets and keeping them for an extended time. Although both approaches have advantages and disadvantages, one key distinction between them is that active investing is far riskier than passive investing.

How are Blockchains used in different sectors?

Blockchain technology has the potential to simplify procedures in a wide range of sectors.

- Blockchain may trace the movement of commodities and materials as they exchange hands in the supply chain business, for instance. As a result, there would be more accountability and openness, and the likelihood of fraud would be lower.
- Blockchain technology can be implemented to the healthcare sector to protect patient information and speed up the billing and claims process.

What characteristics characterize Blockchain technology?

- A distributed ledger that uses Blockchain technology is safe, open, and unchangeable.
- Blockchain technology has the prospective to completely change how we interact with the digital world by enabling the creation of a decentralized database that cannot be altered.
- Blockchain technology is tamper-proof, secure, and transparent.

What constitutes Blockchain technology's core elements?

The three main parts of Blockchain technology are as follows:

- The consensus method, the distributed ledger, and the smart contracts.
- A database dispersed across a network of computers is the distributed ledger. The network of computers can agree on the ledger's current state thanks to the consensus process.

- The Blockchain can be used for more than simply a database, thanks to smart contracts.

What is a Blockchain protocol?

Three of the most popular protocols the Blockchain protocol employed by Bitcoin, which was the first and is now the most popular, is:

- Bitcoin: Often referred to as a cryptocurrency, Bitcoin is a decentralized digital currency. It is based on a decentralized computer network known as a Blockchain that records all currency transactions. Bitcoin uses a proof-of-work technique to authenticate transactions and record them on the Blockchain. The most well-known cryptocurrency is Bitcoin, the first to be invented.
- Ripple: A cryptocurrency comparable to Bitcoin, Ripple uses a distributed computer network to keep track of all money transactions. It validates transactions and adds them to the Blockchain using a proof-of-work method. Founded in 2012, it is the second-largest cryptocurrency by market capitalization.
- Ethereum: In 2013, Vitalik Buterin published a white paper outlining the Ethereum Blockchain. Buterin, a programmer from Canada raised in Russia, has been associated with Bitcoin since the beginning. Although intrigued by the technology, he believed that Bitcoin lacked a scripting language for creating applications. He developed a brand-new system that would be more versatile.

What distinguishes a Blockchain from a database?

A database is centralized if it is managed by just one organization. This entity may be an organization, an authority, or a person. While a Blockchain is decentralized, meaning no organization controls it.

What distinguishes Blockchain from the cloud?

As opposed to the cloud, which is centralized, Blockchain is decentralized. Blockchain is immutable, meaning it cannot be modified once data is added to the Blockchain. As a result, Blockchain is dispersed throughout a network of computers, whereas the cloud is saved on a single server.

What is Blockchain-as-a-Service?

Thanks to the Blockchain-as-a-service cloud-based service, customers can create, host, and use their apps, smart contracts, and other functionalities on the Azure cloud platform. Thanks to Azure's integrated services, Blockchain applications are simple to design, deploy, and administer. Customers can create and deploy Blockchain applications using Azure's managed services without having to set up and manage their infrastructure.

What effects does Blockchain technology have?

On society, Blockchain technology has had a significant impact, including:

- Bitcoin, the primary use of Blockchain technology and the main driver behind its creation, has benefited many through financial services like digital wallets. It has given people in difficult financial situations access to micro-loans and permitted micropayments, reviving the global economy.
- The notion of TRUST is where the following significant change will occur, particularly in the context of multinational business. In the past, lawyers were retained to help two parties who didn't trust one another, but this added time and expense. However, the introduction of cryptocurrency has fundamentally altered the equation for trust.

Numerous organizations are situated in places with limited resources and high levels of corruption. Blockchain offers these impacted individuals and organizations a major edge in these situations by enabling them to avoid the tactics of dubious third-party intermediaries.

- The Internet of Things (IoT) is a brand-new reality already brimming with smart gadgets. It can do anything from turning on your washing machines to operating your cars, maneuvering your ships, planning trash pick-up, controlling traffic safety in your neighborhood, and more! Blockchain is helpful in this situation. By using Blockchain technology to build smart contracts, any business can enhance processes and maintain more precise records in all of these situations (and more).
- Blockchain technology allows businesses or apps like Airbnb and Uber to operate on a decentralized peer-to-peer network. It enables consumers to pay for expenses like parking and tolls.
- The healthcare sector can employ Blockchain technology as a safe platform to store private patient information. The technology allows health-related companies to establish a central database and communicate the data exclusively with those with the necessary permissions.
- Two parties who want to execute a private transaction can use Blockchain technology in the world of private consumers. However, before both parties can move forward with these transactions, several elements need to be worked out:
 - What are the exchange's terms and conditions (T&C)?
 - Are all of the terms understood?
 - When does the conversation begin?
 - When does it end?
 - When is it unfair to end the conversation?

All parties may instantly obtain the answers to these queries by looking up "blocks" in the "chain" since Blockchain technology uses a shared, distributed ledger on a decentralized network. All transactions in a chain can be used to track a transaction on a Blockchain platform from its origin to its final destination.

Applications of Blockchain and Real-World Use Cases

Applications for Blockchain go well beyond cryptocurrencies like Bitcoin. The technology is affecting many industries, ranging from how contracts are enforced to making the government run more effectively. It can increase openness and justice while also saving businesses time and money.

Thirty-four real-world Blockchain application cases for this practical yet ground-breaking technology have been compiled for us. Even though this list is far from complete, they are already altering how we conduct business.

Blockchain in money transfers

Transfer apps for cryptocurrencies are now experiencing an explosion in popularity, led by Bitcoin. Due to the time and money that Blockchain may save financial institutions, it is particularly well-liked in the industry.

Blockchain can save the most prominent banks a ton of money by decreasing third-party fees, eliminating bureaucratic red tape, and making ledger systems real-time. These businesses efficiently move money using Blockchain technology.

1. Chainalysis: location: New York, New York

Chainalysis creates tools to assist governments and financial organizations in keeping an eye on the trading of cryptocurrencies. The business's due diligence software keeps an eye out for and picks up on illegal money laundering, trading, and compliance problems while promoting Blockchain trust.

2. Circle: location: Boston, Massachusetts

Boston-based Circle is in charge of regulating cryptocurrency payments and exchanges between users. Seven other cryptocurrencies are available on Circle's investment and money transfer platform, including Zcash, Monero, and Bitcoin.

3. Algorand: location: Boston, Massachusetts

Algorand creates technology that bridges the divide between centralized and decentralized finance, using cutting-edge goods and protocols to improve the fairness of financial exchanges. The company's technology is made feasible by a collection of layer-1 Blockchains, with use cases ranging from

securities and supply chains to insurance, gaming, digital contracts, and beyond. These Blockchains offer scalability, security, transaction finality, privacy, co-chain, and smart contract capabilities.

4. Chain.io: location: San Francisco, California

Chain creates Blockchain cloud infrastructures for the financial services industry. Financial institutions can move cryptocurrency with safety and efficiency because of the San Francisco company's digital ledgers.

Blockchain smart contracts

Smart contracts are similar to regular contracts, but they eliminate the middleman and offer degrees of accountability for all parties involved that are not feasible with conventional agreements. Instead, the rules of the contract are enforced in real-time on a Blockchain, ensuring compliance from all parties involved while saving businesses time and money.

As industries, including government, healthcare, and real estate, learn about the advantages, Blockchain-based contracts are growing in popularity. Here are a few examples of how businesses utilize Blockchain to enhance contracts.

5. Dfinity: location: San Francisco, California

Users can create decentralized applications and smart contracts with Dfinity's Internet Computer as an open-source Blockchain network, at any file size. Smart contracts can be directly loaded into browsers as their website, saved directly on-chain, and made available without using tokens or wallets by utilizing its network services.

6. BurstIQ: location: Denver, Colorado

Big data Blockchain contracts from BurstIQ enable the secure exchange of sensitive medical data between patients and practitioners. The smart contracts define the conditions under which data can be shared and even show specifics of each patient's individualized treatment plans.

7. Mediachain: location: New York, New York

Using smart contracts, Mediachain pays musicians what they are due. Artists can agree to increased royalties and receive payment in full and on time by engaging in a decentralized, transparent contract. In April 2017, the streaming behemoth Spotify purchased Mediachain.

8. Propy Inc.: location: Palo Alto, California

Global real estate marketplace Propy features a decentralized mechanism

for title registries. The online marketplace offers properties that can be bought using cryptocurrencies and uses Blockchain to make title issues quick.

IoT and Blockchain

The subsequent logical explosion in Blockchain applications is the Internet of Things (IoT). IoT offers a wide range of uses and several security issues. As the number of IoT products rises, hackers will have more opportunities to steal your data via devices ranging from Amazon Alexa to smart thermostats.

With Blockchain infused through technology's transparency and the theoretical incorruptibility to keep things "smart," IoT adds a layer of security to avoid data breaches. The businesses using Blockchain to improve the Internet of Things security and intelligence are listed below.

9. HYPR: location: New York, New York

With its decentralized credential solutions, HYPR reduces the cybersecurity concerns associated with IoT devices. The company practically impenetrable IoT devices by removing passwords from a central server and employing biometric and password-free solutions.

10. Xage Security: location: Palo Alto, California

The first cybersecurity platform for IoT enterprises in the world is called Xage. The technology can even self-diagnose and repair potential breaches while managing billions of devices simultaneously. IoT businesses in the manufacturing, energy, and transportation sectors frequently use Xage.

11. Helium: location: San Francisco, California

The largest continuous wireless network in the world that connects and exchanges data across the Internet of Things devices is Helium's People's Network. It is built on Blockchain to support monitoring and management initiatives for smart agriculture, smart cities, smart water, and logistics. The network can track and report real-time asset data. To mine HNT cryptocurrency, hotspot devices that work with the People's Network can be employed.

Blockchain security

Up to 42 million Americans may have become victims of identity fraud in

2021 alone, according to AARP. Fraud on this scale can take many forms, from falsifying documents to accessing private data. The government may see a sharp decline in identity theft claims by storing social security numbers, birth dates, birth certificates, and other sensitive information on a decentralized Blockchain ledger. Here are a few companies that use Blockchain technology in identity security.

12. Ligerio: location: Rochester, New York

A powerful platform for enabling decentralized cooperation both on and off Blockchain, Ligerio offers lightweight, expandable protocols for secured multiparty computation and zero-knowledge proofs. Users can conduct private transactions, smart private contracts, safe auctions for decentralized exchanges, and activate verified machine learning capabilities on the platform, which is managed by a team of finance and cryptography experts with years of experience.

13. Civic: location: Palo Alto, California

Civic is a Blockchain-based ecosystem that allows users to see who has access to their data. Users of the company enter into smart contracts where they specify who and how much of their personal information can be shared. The person is promptly informed if the agreement is breached or if an unauthorized source tries to access confidential data.

14. Evernym, Inc.: location: Salt Lake City, Utah

Using distributed ledger technology, the Sovrin identity ecosystem from Evernym enables users to maintain their identities throughout the web. Sovrin maintains confidential information, serves as a conduit for communication between the person and others interested in confidential information, and instantly validates the accuracy of information.

15. Ocular: location: Los Angeles, California

Ocular's anti-money laundering compliance platform uses Blockchain-enabled security to ensure that data cannot be altered. The technique uses biometric systems to scan applicants' faces for passports, licenses, and other official identification documents. Governments can more readily capture identity thieves using forged passports, certificates, and IDs from other nations by viewing biometric systems on Blockchains.

Blockchain in healthcare

Despite its early usage, Blockchain in healthcare is already showing some

potential. Early Blockchain systems have demonstrated the ability to lower healthcare expenses, increase stakeholder access to information, and simplify business procedures. Ensuring that an already bloated sector can reduce extravagant costs, an innovative system for gathering and sharing private information may be exactly what the doctor ordered.

16. Chronicled: location: San Francisco, California

Using Blockchain, chronicled automated product verification and commercial transactions in the life sciences sector. The business maintains DSCSA compliance for distributed pharmaceutical products and stores records of contracts between manufacturers and wholesalers in its MediLedger network.

17. WholeCare: location: Baltimore, Maryland

WholeCare streamlines and simplifies the influx of information required to effectively manage and provide care for individuals who need it when they need it most for both healthcare professionals and families giving care. Using the WholeCare platform, individuals, support networks, and multi-care facilities can better understand how to deliver the most nuanced care possible—combining information about care plans, medication protocols, appointment scheduling, and high-quality resources into one easily distillable platform. WholeCare offers HIPAA-compliant record keeping so everyone participating in the care process may stay at ease. It is built on a Blockchain network to remain as safe as possible.

18. Patientory: location: Atlanta, Georgia

Blockchain technology underpins Patientory, an all-in-one medical record system for patients and physicians. The dispersion of data among many providers and clinics is one of the main problems facing healthcare. Patientory provides continual, secure access to a patient's medical history, records, current providers, and everything else a doctor could need to know. Any medical practitioner may swiftly and safely diagnose patients based on a more precise medical history thanks to the Blockchain platform, which enables constant contact between patients and doctors.

19. Nebula Genomics: location: San Francisco, California

Nebula Genomics is on a quest to understand the human genome better and increase the accessibility and affordability of personal genomics. Only the company's whole-genome DNA sequencing tests can decode a person's DNA. Because every data acquired from a specific test is fully anonymous and kept secret using Blockchain-based encryption, a user's data can never be

traced back to them or stolen.

20. Medicalchain: location: London, England

Both patients and medical professionals benefit from the easier and more secure information flow made possible by Medicalchain's collaborative Blockchain platform. Doctors no longer need to wait for insurance information, thanks to Medicalchain, and the Blockchain can automatically check if a patient is covered by insurance. A Blockchain-based portal that safely displays patient medical records and identifies top prospects for various trials can also be used to discover top candidates for drug and clinical trials quickly.

Blockchain logistics

A significant issue in the shipping sector is the need for more transparency and communication brought on by the dense concentration of logistics firms. Over 500,000 shipping companies operate in the US, which leads to data siloing and transparency problems, according to a joint study by Accenture and logistics behemoth DHL. According to the paper, supply chain management and logistics are plagued by several issues that can be resolved by Blockchain technology.

Furthermore, Blockchain creates data transparency by making a single source of truth visible. Blockchain can increase trust in the sector by recognizing data sources. Technology has the potential to save the logistics sector billions of dollars annually by streamlining and automating the process. Blockchain is a safe and affordable alternative for the logistics sector. These businesses are at the forefront of Blockchain technology in logistics.

21. Slync.io: location: Dallas, Texas

Slync.io's Logistics Orchestration platform uses Blockchain and AI technology to give shipping accountability and analytics. Access to a collaborative business network, document management, and order automation are all available to simplify logistical operations.

22. DHL: location: Plantation, Florida

The world's largest shipping company, DHL, is at the vanguard of Blockchain-based logistics, employing technology to track shipments digitally and protect the integrity of transactions. One of the biggest transportation businesses to adopt Blockchain is DHL, which is well-known

in the US.

23. Maersk: location: Florham Park, New Jersey

Shipping behemoth Maersk, headquartered in Denmark with branches across the US, has partnered with technology behemoth IBM to integrate Blockchain into international trade. Blockchain will be used by the two businesses to track items digitally across international borders in real-time and gain a better understanding of supply chains.

24. ShipChain: location: Los Angeles, California

A fully integrated Blockchain solution that supports the entire shipping process is called ShipChain. The logistics ecosystem safely tracks and records each step of the shipment's journey from the moment it leaves the facility until it reaches its destination, building a transparent ledger. Los Angeles-based ShipChain wants Blockchain to revolutionize the \$8.1 trillion supply chain business.

Blockchain and NFTs

Since cryptocurrencies, non-fungible tokens (NFTs) have been the most popular Blockchain application. These digital products, already sweeping the globe, began becoming more prevalent in 2021. NFTs are digital goods sold on a Blockchain, allowing a single owner to assert full ownership rights, such as music, art, GIFs, and films. Blockchain technology will enable consumers to claim exclusive ownership over some of the most sought-after digital products.

Recall the 2011 internet meme Nyan Cat? On the Blockchain, that famous GIF was sold for \$600,000 in Ethereum. The digital artist "Beeple" has never made a sale exceeding \$100 before October. His digital piece, The First 5000 Days, sold in March 2020, fetched an astonishing \$69 million. With NFTs, consumers can own digital memories, works of art, and cultural artifacts that will outlive us all.

A few instances of businesses profiting from the NFT trend.

25. RECUR: location: remote

RECUR turns popular IPs and brand icons into digital treasures for consumers. Fans of movies or sports can purchase and resell NFTs associated

with their preferred franchise and take part in tailored challenges to improve their marketplace experience. The services of RECUR have been utilized by Paramount, collegiate basketball teams, and others to develop NFT marketplaces for IPs.

26. Candy: location: New York, New York

A secure marketplace where users may purchase, sell, and trade NFTs to optimize their investment is made possible by Candy's NFT ecosystem. It enables fans and collectors to interact with prominent sports, art, music, and cultural icons as officially sanctioned digital collectibles. To make their NFT projects a reality, the organization works closely with athletes, creators, and content owners, and one such series highlights the 30 Major League Baseball venues.

27. Dapper Labs: location: remote

Dapper Labs is one of the first businesses to take off due to the NFT frenzy. It collaborated with the NBA to create "NBA Top Shot," an NFT marketplace where customers may purchase digital content from their preferred NBA players or teams. Customers can buy memorable moments like a highlight-reel slam by LeBron James or a GIF of Anthony Davis blocking a shot. Top Shot has generated more than \$500 million in sales so far.

28. Pixura: location: Remote

A software called Pixura enables non-technical people to develop, monitor, and trade digital NFTs on the Blockchain. The business allows everyone to quickly produce digital assets, from game studios to individual musicians and artists. Additionally, the company founded the website SuperRare, which in the NFT age, has emerged as a top destination for buying and selling digital art.

Blockchain in government

Blockchain technology may have some unexpected uses in bettering the government. As previously stated, certain state governments, such as Illinois, are already utilizing the technology to secure official records. Still, Blockchain technology may also increase bureaucratic efficiency and accountability and lessen astronomical budgetary burdens. According to the New York Times, Blockchain can reduce millions of hours of bureaucracy annually, hold government officials responsible through smart contracts, and

promote transparency by keeping a public record of every activity.

Blockchain may revolutionize the electoral process. Voting on mobile devices is made possible by the security and incorruptibility offered by Blockchain-based voting, which could increase civic involvement.

A few examples of how businesses and organizations use Blockchain technologies to improve government.

29. Kaleido: location: Raleigh, North Carolina

Kaleido claims to provide solutions for managing and deploying ledgers using Blockchain technology in seven different businesses, including the government. The company's services can be used for audits, risk reduction, and general operations streamlining in the public health, education, customs, and payment sectors. Organizations, including the World Wide Fund for Nature, the Centers for Disease Control and Prevention, and the United Nations have adopted Kaleido's solutions.

30. Vatz: location: Boston, Massachusetts

Voatz is a Blockchain-based mobile voting platform. Voting on a mobile device from anywhere globally is secure thanks to the encrypted biometric security technology, which prevents data corruption and hacking. The state of West Virginia was amidst the first to use the platform to gather votes from overseas citizens and active-duty military people.

31. Follow My Vote: location: Blacksburg, Virginia

Follow My Vote is an open-source virtual Blockchain voting system that is safe for use online. The technology allows access from any device and reduces the cost of purchasing paper votes. Follow My Vote offers the end-to-end tools required for elections to ensure voter safety and confidence.

Blockchain in the media

Data privacy, royalties, and intellectual property infringement are current issues in the media. According to a Deloitte report, the widespread sharing of content that violates copyrights has been facilitated by media digitization. Deloitte thinks Blockchain can give the sector a much-needed facelift regarding data rights, piracy, and payments.

The advantage of Blockchain in the media sector is its capacity to stop

digital assets like mp3 files from existing in many locations. Through the use of a transparent ledger system, it can be shared and distributed while also maintaining ownership, making piracy all but impossible.

Additionally, Blockchain can protect data integrity, enabling marketing departments to target the right consumers and musicians to be paid appropriately for their unique compositions. The following businesses are promoting Blockchain in the media at an increasing rate.

32. MadHive: location: New York, New York

Digital marketers can use MadHive, Blockchain-based advertising, and data solutions. The platform monitors, archives, and produces reports on user activities while storing all the information on a personal Blockchain. Advertisers may gain insights into their clients without compromising data privacy thanks to MadHive's tailored audience reports and real-time data monitoring.

33. Steem: location: Austin, Texas

A social media site powered by Blockchain is called Steem. Tokens are used as incentives in its "Proof-of-Brain" community to motivate users to produce original content. The number of upvotes any article receives determines how many tokens are given out. Over \$50 million in tokens have been paid to creators by Steem.

34. Open Music Initiative: location: Boston, Massachusetts

A nonprofit organization, Open Music, is developing an open-source system to identify music rights owners. The organization is making it simpler for artists and musicians to be recognized for their work and fairly compensated by entrusting their music rights data to the Blockchain. Media behemoths like Netflix and Spotify, as well as other segments of the music industry, including producers and radio stations, support the idea.

Key Features in This Chapter

- In terms of technology, Blockchain is a digital ledger that has recently attracted much interest.
- Customers can create, host, and use their Blockchain apps, smart contracts, and other functionalities on the Azure cloud platform thanks to the Blockchain-as-a-service cloud-based service.
- Two parties who want to execute a private transaction can use Blockchain technology in the world of private consumers.
- There are 34 real-world Blockchain application cases for this practical, yet ground-breaking technology have been compiled for us.
- Businesses are using Blockchain to improve the Internet of Things security and intelligence.
- Over 500,000 shipping companies operate in the US, which leads to data siloing and transparency problems, according to a joint study by Accenture and logistics behemoth DHL.
- Deloitte thinks Blockchain can give the sector a much-needed facelift regarding data rights, piracy, and payments.

THE BLOCKCHAIN AFTER THE FTX CRASH

Investor trust has decreased as a result of several recent market disasters. Cryptocurrencies were entering the mainstream as 2022 got underway. Leading institutional investors had made large bets in cryptocurrency the year before, giving the market credibility.

Bitcoin and ether values have already dropped by around 75% from their all-time highs due to a slew of industry failures, including the collapse of Bahamas-based FTX Trading Ltd. in November. Financial advisors and cryptocurrency specialists are split on the future of the sector.

Many crypto doubters have interpreted the demise of FTX as confirmation of their darkest fears. According to Andrew Pyle, a financial advisor at CIBC Wood Gundy in Peterborough, Ontario, "[Cryptocurrency] is highly speculative, highly volatile, and it's something you steer clear from until we see signs of maturity in that area."

According to Pyle, who has never advised investing in cryptocurrencies, the sector has no place in the portfolios of the "overwhelming majority" of clients today, even though the Blockchain technology that powers cryptocurrencies is likely to bring investment opportunities in the future.

Mark Noble, executive vice-president of ETF strategy at Horizons ETFs Management (Canada) Inc., stated that "you're looking at several years before you gain broader public faith in cryptocurrencies" as a result of the impact of FTX's collapse.

That said, Noble argued that Blockchain advancements would continue even if there were a period of low investor interest known as the "crypto winter." Both long and short bitcoin funds are available through Horizons

ETFs.

Consistent with Michael Zagari, a Montreal-based financial advisor with Burlington, Ontario-based Mandeville Private Client Inc., institutional investors who believe in Blockchain technology's transformative potential are likely to "double down" on their investments. There will be "a lot of bargains to be had," according to Zagari, who expects the business will remain unpredictable for the next 12 to 18 months even though "just because one investment [FTX] didn't perform doesn't mean they're going to stop there."

For risk-tolerant customers with at least a ten-year time horizon, he advises investing no more than 10% of a portfolio in bitcoin or investments related to cryptocurrencies as part of their equity exposure. Other cryptocurrency exchanges have had to halt withdrawals due to FTX's insolvency, and others are battling to stay in business.

Additionally, "additional shoes are about to drop. We just don't know how many or how big," said Alex Tapscott, managing director of Ninepoint Partners LP's digital asset group, during a webcast on November 24.

The Ontario Teachers' Pension Plan Board and the Caisse de dépôt et placement du Québec exposed themselves to "huge concentration risk" by investing in a single crypto firm rather than taking positions in well-established cryptocurrencies, as claimed by Tapscott, who describes himself as "short-term bearish, long-term bullish" on cryptocurrencies.

Bitcoin and ether will survive this fall, based on Brian Mosoff, CEO of Toronto-based Ether Capital Corp., even if some exchanges don't. The technology hasn't altered, and the principles and value proposition of bitcoin and ether "still do exactly what they set out to do."

The general public concurs. According to National Bank Financial Markets data, only \$66 million of the loss in crypto ETF assets under management between January 1st and November 30th was caused by withdrawals (NBFM).

As mentioned by Daniel Straus, director of ETF research and financial products research with NBFM, "it seems like the crypto ETF users in Canada are keeping to their allocations," in an email to Investment Executive. The anemic outflows from Canadian crypto ETFs indicate its investors may consider them like "moonshot" long-term bets, even though bitcoin and ether are both exceedingly hazardous and speculative.

After the failure of FTX, Mike Tropeano, Senior Director of Wealth Consulting at Broadridge Financial Solutions Inc. in Boston, predicted that

international regulators would "be much harsher" toward the cryptocurrency business. The most well-known names will flee for safety in the upcoming years, and there will be a "thinning of the herd" and an increase in inventiveness.

Tropeano stated that "the information is still pouring every day, [not only] regarding FTX, [but also] with the overall market." "Anyone wishing to participate in the market — the [financial] advisor especially — needs to exercise great attention to stay current on events,"

In the opinion of Mosoff, some "advisors are probably relaxing a little bit" because they know that their clients are less likely to inquire about cryptocurrency investments "until the next cycle starts." However, he predicted that the cycle would occur and advised advisors to use the crypto winter to further their education.

Although the market has been shaken by the failure of FTX and BlockFi Lending LLC, the severity of the crypto winter may depend on how the sector's stalwarts fare under pressure, as believed by Daniel Gonzalez, research analyst and consultant in Toronto for the California-based Javelin Strategy & Research: "I believe that the adoption of cryptocurrencies by retail investors would be halted for a period if a Coinbase, crypto.com, or Binance were to go under."

The advisor's job today is similar to what it was during the height of cryptocurrency trading in many aspects. There is unquestionably a role for advisors to play in this situation, and that duty is to speak with reason when it comes to investing in cryptocurrencies, according to Mosoff. However, I don't believe that means "I'm going to rule out an asset class completely."

Despite this, according to Jason Heath, managing director of Objective Financial Partners Inc. in Markham, Ontario, cryptocurrencies have not shown to be a hedge against inflation or a diversifier. Furthermore, "Speculative investments like cryptocurrencies in 2023 are expected to be hampered by increased interest rates and a projected recession."

If institutional investors, traders, analysts, portfolio managers, hedge fund managers, and even regulators can't grasp this space, Pyle claimed that it is unfair to expect individual investors to do so.

Protecting a client's wealth is an advisor's "first duty," Pyle said. "Protect it de facto from things that most people don't understand, prevent it from inflation, and protect it from running out."

Highs and Lows in the Crypto Space

2021

February 18: The first bitcoin ETF is released by Purpose Investments Inc. of Toronto.

October: Through its Teachers' Venture Growth platform, the Ontario Teachers' Pension Plan Board (OTPP) invests US\$75 million in FTX Trading Ltd.

October 12: Celsius Network LLC, a cryptocurrency lender, based in New Jersey, receives a US\$150 million investment from Caisse de dépôt et placement du Québec.

November 10: The all-time intraday high for bitcoin is US\$68,789.63.

November 16: At \$4,891.70, ether reaches its all-time intraday high.

2022

January: The OTPP continues its \$20 million US investment in FTX.

January 10: Fidelity Investments Canada ULC announced that its asset-allocation ETFs will now include 1% to 3% exposure to bitcoin. (At the time of this publication, the target allocations are unchanged.)

February 14: New Jersey-based BlockFi Lending LLC agrees to pay \$100 million in fines to the US Securities and Exchange Commission and pursue registration of its cryptocurrency lending service.

March 31: According to information from National Bank Financial, Canadian cryptocurrency ETFs have \$6.1 billion in assets under management.

May: The linked token Luna and the stablecoin TerraUSD both crash. Bitcoin fell below US\$25,000 on June 18. Early in May, it was trading at much more than \$50,000.

July 6: A crypto broker with headquarters in New York, Voyager Digital Ltd., files for Chapter 11 bankruptcy.

July 13: Celsius Network declares bankruptcy under Chapter 11.

August 17: Announcing the write-down of its entire investment in Celsius Network, Caisse de dépôt et placement du Québec.

November 11: FTX declares bankruptcy under Chapter 11.

November 17: OTPP declares that it will write off all of its FTX stakes.

November 21: Senators Elizabeth Warren, Dick Durbin, and Tina Smith write to FMR LLC (Fidelity Investments) requesting that the company rethink its decision to permit access to bitcoin in 401(k) plans.

November 27: A US cryptocurrency exchange called Bitfront has announced that it will shut down in March 2023. The decision "is unrelated to recent difficulties related to certain deals that have been accused of misbehavior," according to the platform.

November 28: BlockFi declares bankruptcy under Chapter 11, alleging exposure to FTX.

What Will Happen to Cryptocurrency After FTX Collapses?

Does the cryptocurrency market still have a bright future, or is the cryptocurrency bubble about to burst? A recent uptick in bankruptcy filings within the cryptocurrency industry has sparked worries about the long-term sustainability of digital assets and the industry.

First, think about the extent of the carnage: there has been a remarkable rise in the number of cryptocurrency market bankruptcy attributed to liquidity issues. Since their businesses could not continue, Three Arrows Capital, Alameda Research, Voyager Digital, FTX, Genesis BlockFi, and Celsius Network have either halted customer withdrawals or filed for bankruptcy.

The collapse of FTX, previously one of the biggest cryptocurrency exchanges in the world, was caused by a lack of assets on its balance sheet. Customers withdrew \$650 million in assets on November 7 in response to rumors that the exchange may need to be properly liquidized. As a result, it was discovered that FTX was using customer funds to finance dangerous bets made by the affiliated trading company Alameda Research. The day before it failed, the exchange had only \$900 million in liquid assets to offset its \$9 billion in liabilities. The collapse of FTX, a market leader in the cryptocurrency space, was a big shock and diminished trust in digital assets.

Bitcoin owners have suffered significant losses: When you sum up the fall in the value of bitcoin, ether, and all the other digital currencies since they were at or around their peaks last year, you end up with a loss of more than \$2 trillion in market cap.

Most of the current issues are on crypto lenders, an area of the cryptocurrency business that has experienced rapid growth in recent years. Crypto users who deposit their digital assets at a crypto lender or digital exchange also make money, just like customers of conventional banks do with interest rates on their savings. Due to historically low-interest rates, savings accounts at banks only provided minimal returns over the past couple of years. However, certain cryptocurrency lenders and exchanges provided substantially greater yields, frequently in the double digits and occasionally as high as 20%.

The truth about the unsustainable yields provided by some cryptocurrency exchanges and lenders has come to light. These cryptocurrency companies, like banks, make money from deposits by lending.

The fees that borrowers pay for the loan are a percentage, and cryptocurrency lenders earn from the difference between the interest payments made to depositors and the fees paid by borrowers. However, unlike conventional regulated lenders, crypto firms aren't under the control of banking regulators; as a result, few regulations are governing the capital they must retain or the things they may do with their clients' digital assets.

The current failure of these cryptocurrency lenders has also highlighted how intertwined many of these businesses are. FTX gave affiliated trading company Alameda Research billions of dollars, which were then used to finance dangerous trades. Alameda Research and Three Arrows Capital were on the list of debtors for the cryptocurrency lender BlockFi.

More cryptocurrency business failures appear likely. Additionally, other exchanges besides FTX were involved in shady trading, utilizing depositors' money when customers became concerned about the security of their digital currency deposits and wanted their money back.

Who is most at risk to the effects of crypto?

Thankfully, banks are secure. They each hold roughly \$9 billion in cryptocurrencies. Banks paid attention to regulators' warnings to be cautious while investing in the asset class. Unexpectedly many retail investors possess cryptocurrency and experience some losses. Although broad, that exposure is very brief. In the US and Europe, 10% of homes have cryptocurrencies. The typical holding in the US is \$1,000, and most European investors hold under \$5,000.

Venture capital firms, which have staked much money on the cryptocurrency market, are the true bag holders. Only a few hundred of the 10,000 or so companies connected to the cryptocurrency industry are publicly traded. Venture capitalists, who funded the majority of the remainder, are now facing significant (and, in some cases, almost entirely) losses.

So, is the once-booming industry's end near?

Not exactly. Some more popular currencies, such as bitcoin and ether, appear resilient. For instance, Bitcoin is still four times higher than it was in December 2018 despite a 75% decline from its top in November 2021. The second-placed token, ether, has increased by more than 1,000%. But there is unquestionably a significant shrinking happening. Not all tokens will survive. Since their customers prefer to keep their cryptocurrency in "digital wallets," which often don't allow holders to earn a return on their cryptocurrency but ensure their assets stay safe, many exchanges and crypto lenders won't.

The exchanges that endure will have to put in a lot of effort to demonstrate to users that their money is safe and liquid rather than being the next FTX waiting to crash.

Washington will likely impose more regulations in the future to put an end to the cryptocurrency industry's ad hoc character. This shake-up is necessary for the industry's survival to eliminate the currencies and businesses that are nothing more than hype and to focus on the few promising players. It will be much tougher to demonstrate their usefulness and practical value now; gone are the days of easy riches.

Learn From FTX And Stop Making Speculative Investments

Speculative investing drew people to FTX. Now that we have seen the results let's hope they start seeking value-based propositions. The FTX crash highlights more than just another Bitcoin exchange going down. It serves as a reminder that the sector must develop and accede to value; this is where the value gap is.

FTX is ranked as the second-biggest cryptocurrency exchange globally. It now serves as a metaphor for the death rattle caused by the massive amounts of money invested in modernized, centralized corporate models that have been given the appearance of fake decentralization.

According to legendary investor, Warren Buffet, you only discover who has been swimming naked when the tide goes out. Some nudists participated in the most recent round. But this has undoubtedly happened before. Actually, not quite.

Bitcoin BTC was down \$16,744 at the start of the longest financial market bull run in history. The industry it gave rise to almost exploded under the best of conditions. But even good things come to an end. Regulators seeking control and deteriorating macroeconomic conditions have produced an unfavorable convergence for the bitcoin business.

In the meantime, value-based investing has returned to traditional markets with caution. Simple: money was practically free when interest rates were at historic lows. Not right now. The meteoric ascent of Uber, Airbnb, and DoorDash was made possible because businesses that made money were not appreciated when it was offered for free. However, today's promises still need to be made. Investors will need value confirmation before putting up their progressively more expensive capital.

Value-driven investing will be made possible in crypto markets for the first time with the demise of FTX. FTX Token FTT is down \$1.06, demonstrating the falsity of tokenomics. And economics is, despite how often we ignore its teachings when the economy is booming. There is both supply and demand. When markets are in equilibrium, they can function. If they are not, markets do not function.

We already know that centralized control of bitcoin markets could be more helpful. There are far too many opportunities for money-hungry con artists to exploit people unfamiliar with sophisticated technologies. What

happened? Those who believed in a pot of gold at the end of the cryptocurrency rainbow had their hopes shattered. But among the chaos, the value split offers a glimpse of hope.

What is the value schism?

According to industry jargon, crypto is experiencing a "hard fork." Those left standing after the FTX dust has settled have two options: they keep placing blind bets to discover a "greater fool," or they hunt for information that can be gathered and provided to users.

Some will stick with the latter path. Poor habits continue. However, as investors want more, they will vanish. Initiatives on the Web 3 that return to traditional commerce and offer real value will gain favor in the interim. Successful people will be rewarded handsomely. The end will come swiftly for those who are merely providing the same old cheerleading from the past.

Navigating a new paradigm

There are two principles to consider within the value schism. First, cryptocurrencies are referred to be financial assets. Second, Blockchain is referred to as a framework for technology.

It is challenging to evaluate cryptocurrencies as a class of financial assets due to the necessity for a functional model for pricing protocols. This is to be anticipated in a startup company. In the beginning, there were no criteria by which to evaluate these networks. Retrofitted markets were built for matured markets.

Cryptography has altered since then. We can categorize networks now that we have a fundamental understanding of the numerous uses for decentralized finance (DeFi) protocols. The proof-of-work chain used by Bitcoin is widely spread, slow, and secure. It is possible to see how many Bitcoin wallets there are and how they interact with the network. The secondary transaction layer, the Lightning Network, allows for calculating the value delivered across it.

Ethereum is a proof-of-stake chain. Despite being more centralized than Bitcoin, it is the foundation of DeFi. DeFi developed a technology called total-value-locked computations to help in value estimation. Even if they

have some limits, creating sophisticated financial gauges outside established institutions is quite interesting. Traditional finance concurs, which explains why regulatory attention is increasing.

In 2016, it seemed similar to selling Bitcoin or Ether for \$1,187. Now that these networks are more distinct, we can assess them using a wide range of data-driven measures. Cryptocurrency is becoming a verified asset class.

The Rise of Functionals

Functionals are non-financial Web 3.0 assets, such as Blockchain products and services. Take ZK (zero knowledge) Proof, for instance. A potential homeowner wants to show a real estate agent that they have the money they need without giving away their account's specifics. They can pay for this service with a ZK. Instead of exchanging or keeping assets, they pay solely for a service that safeguards their anonymity.

Many new emerging data handling initiatives provide services like identification tools, cloud storage, and search and indexing. Thanks to their decentralized infrastructure, they are relatively reasonably priced compared to their centralized competitors.

The failure of FTX is neither extraordinary nor decisive, and the system's infection is complicated because of downward pressures imposed on macroeconomic forces. After all, is said and done; however, FTX will emerge as a growth ring in the cryptocurrency story, demonstrating that a fire passed through and left behind protected systems that will enhance value.

The value split will compel Blockchain ecosystems to go in one of two directions: either keep relying on hype cycles to earn speculative gains or create models that show real user value. Like personal computers matured, Blockchain-based technologies are now spreading from hobbyists' garages to offices and individual wallets worldwide.

Why the Market Will Recover and What to Buy

As 2022 draws to a close, FTX failures and a decrease in the crypto market are giving cryptocurrency traders worldwide the worst fears. However, if we look at the forecasts for 2023, we may conclude that the market will rebound and that now is the ideal time to invest. The following are forecasts for the coming year and promising cryptocurrencies that will take off:

- Dash 2 Trade - the top cryptocurrency to buy right now,
- RobotEra - a potential play-to-earn NFT game,
- Impt.io - a market for carbon credits on the Blockchain,
- Calvaria - a free online card combat game and entryway to the crypto sphere.

Five predictions that point to a recovery of the crypto market

Following are five expert forecasts and analyses that show 2023 will be a bullish year:

- 2023 should be a successful year, according to the cryptocurrency market's historical record.
- The beginning of 2023 will mark the end of the bear market,
- A bull market will begin once the cryptocurrency market reaches its bottom.
- Bitcoin will return.
- Web 3.0 will take the world by storm.

Dash 2 Trade is now the cryptocurrency the greatest to invest in.

The wise investment will be to purchase coins when they are cheap before the market recovers and enters the bull run. Dash 2 Trade is the project that traders are currently most interested in. This platform for crypto analytics

is enhanced with various features that let users access information about the coins they want to buy. The platform's native coin, D2T, is presently up for pre-sale. If you purchase it now, the price will be more than reasonable, and you can count on making money as soon as the pre-sale is over.

Currently, in pre-sale, RobotEra is a promising NFT play-to-earn game.

Users can play as virtual robots in RobotEra, a new NFT multiverse project, on the post-apocalyptic planet TARO. Using the game's native token, TARO, players can develop their robots, enter other worlds, buy land, and restore the planet. Investors have shown tremendous interest in the TARO project, which is currently up for pre-sale. So now is your time to buy cheap coins if you so desire.

Impt.io is a marketplace for carbon credits on the Blockchain.

Impt.io is a promising cryptocurrency initiative linking more than 10,000 businesses and brands. Users can purchase, sell, or burn their NFT carbon credits on this platform and marketplace for carbon credits, thereby lowering their carbon footprint. IMPT, the platform's native token, will be utilized for mining NFT carbon credits. Join the coin today before the market experiences a bull run, as it is currently through a very successful pre-sale.

Free battle card game, Calvaria, is a doorway to the crypto world.

A free and open to all battle card game is called Calvaria. By purchasing NFT collection cards, players can join Calvaria's alternative play-to-earn version. People who want to learn more about play-to-earn games and the cryptocurrency industry should try this game out. Purchase it today before it's too late since the game's native coin, RIA, is also up for presale.

Predictions

Prediction 1: 2023 will be a good year, according to the crypto market's previous performance. In this four-year cycle, the market exhibits bullish years, correction years (during which it declines), decent years, and finally, great years. While 2022 was a year of crypto cold, 2021 was a bullish year. This predicts an excellent year in 2023 will be followed by an even better year.

Prediction 2: At the start of 2023, the bear market will be over.

Some experts believe the crypto market is already declining and will likely reach its bottom by the end of 2022. These forecasts are supported by the cryptocurrency market's historical performance, which indicates that the bear market will conclude within the next three months. This implies that the market should start to climb again at the start of the following year.

Prediction 3: A bull market will emerge once the cryptocurrency market reaches its bottom.

There is no reason why this wouldn't happen again. Every time the cryptocurrency market experienced a bear market, it was followed by a bull run market. Despite the dread gripping investors, those who can persevere or even invest now will be rewarded when the market inevitably erupts.

Prediction 4: Bitcoin will reappear

As previously indicated, experts have observed that BTC always performs in a four-year cycle, and there is no doubt that BTC's performance will have an impact on the entire market. The chart below shows that although BTC experiences ups and downs, it always bounces back. CryptoPredictions predicts that the price of BTC will increase by 2% to start in 2023 and keep rising throughout the year.

Prediction 5: Web 3.0 will become the newest craze.

While most forecasts are based on the industry's historical performance, some cutting-edge developments like Web 3.0 and the decentralization of the cryptocurrency market may project a larger bull run than many analysts anticipated. Users will be able to transact considerably more quickly, and the cryptocurrency market will be significantly impacted. It would help if you kept track of this since it might be the next big thing.

The crypto market will rise

Without sure, the crypto winter will come to an end, and 2023 is likely the year when it does. You will undoubtedly earn if you invest right now in a low-cost, viable project, particularly if you buy coins like D2T, TARO, IMPT, and RIA, which are currently on pre-sale.

Key Features in this Chapter

- Bitcoin and ether values have already dropped by around 75% from their all-time highs a year ago due to a slew of industry failures, including the collapse of Bahamas-based FTX Trading Ltd. in November. Financial advisors and cryptocurrency specialists are split on the future of the sector.
- According to data from National Bank Financial Markets, only \$66 million of the loss in crypto ETF assets under management between January 1 and November 30 was caused by withdrawals (NBFM).
- Value-driven investing will be made possible in crypto markets for the first time with the demise of FTX.
- Impt.io is a marketplace for carbon credits on the Blockchain.
- CryptoPredictions predicts that the price of BTC will increase by 2% to start in 2023 and keep rising throughout the year.

HOW TO BUY STOCKS IN BLOCKCHAIN INDUSTRIES

Discover the benefits of investing in Blockchain stocks and how to get started. Although cryptocurrency is the most well-known use of Blockchain technology, it has much more use than just virtual money. Businesses use Blockchain technology to enhance operations, particularly in decentralized and complicated systems.

Some steps for investing in Blockchain technology and some things to think about.

Quick refresher: what is Blockchain?

Build up your knowledge of Blockchain's operation before investing in it. Blockchain is a public ledger that stores transactional data digitally. Digitally validated and given a different hash (or identity), each "block" of data is then added to the public ledger. In the case of Blockchain-based cryptocurrencies, adding a new block creates a new unit (or coin) of currency.

Once more, Blockchain's capabilities go far beyond virtual currencies. For instance, Walmart (NYSE: WMT) has been experimenting with Blockchain technology to track food distribution from its numerous suppliers. This could make it simpler to identify outbreaks of foodborne illness. Starbucks (NASDAQ: SBUX) has tested a Blockchain project with its coffee growers in Rwanda, Colombia, and Costa Rica to increase pricing transparency.

Why Begin Investing in Blockchain?

The investment community is naturally becoming interested in Blockchain as a new technology that has the potential to have a deep impact on the commercial sector. Several elements that make it appealing are as follows:

- Blockchain can potentially increase an organization's efficiency and profitability over time.
- Prominent tech companies like Salesforce.com and Amazon (NASDAQ: AMZN) are paying attention to the Blockchain (NYSE: CRM).
- The world is rapidly moving to digital thanks to COVID-19. Blockchain is closely related to other ancillary technologies like cloud computing, e-commerce, and artificial intelligence.

In particular, for Blockchain investments, including cryptocurrencies, there are risks to take into account:

- There are numerous new cryptocurrencies with Blockchain initiatives at their core, and many fail to take off.
- The value of cryptocurrencies can fluctuate greatly, and investing in them may result in principal loss.

Ways to Start Investing in Blockchain

There are other ways to participate in the Blockchain market besides purchasing stocks of companies directly.

1. Buy shares of a cryptocurrency trust like Grayscale Bitcoin Trust or directly purchase cryptocurrencies like Bitcoin or Ethereum (OTC:GBTC).
2. Invest in an exchange-traded fund (ETF) that focuses on purchasing shares of businesses exposed to Blockchain technology. Amplify Transformational Data Sharing ETF (NYSEMKT:BLOK) and Reality Shares Nasdaq NextGen Economy ETF are two significant instances (NASDAQ:BLCN).
3. Purchase a new cryptocurrency issued by a developer working on a new Blockchain project as part of an initial coin offering (ICO) to participate in financing a new cryptocurrency.
4. Putting money into publicly traded Blockchain firms..

Then there's the possibility to buy stock in businesses using or developing Blockchain technology, like Starbucks or Walmart. A corporation can become leaner and more profitable by implementing a digital ledger system; over time, higher earnings equate to higher share values.

However, other businesses are placing more specific wagers on the Blockchain. PayPal Holdings (NASDAQ:PYPL), a leader in digital payments, permits businesses to accept bitcoin payments through its Braintree subsidiary. Additionally, the company is developing additional ways to include Blockchain and cryptocurrency buying and selling functionality in its PayPal and Venmo digital wallet apps.

Similarly, Bitcoin may be bought and sold using Square's (NYSE:SQ) Cash App digital wallet. Established digital payments corporations Visa (NYSE:V) and Mastercard (NYSE:MA) are also collaborating with cryptocurrency and Blockchain startups to keep their payment networks relevant as times change.

Leader in commodities and financial derivatives exchanges, CME Group (NASDAQ:CME) is also noteworthy for launching the first bitcoin futures

and options market.

Additionally, Facebook (NASDAQ:FB) is still working on integrating financial services and digital payments into its apps through its Libra project. Government regulators have repeatedly thwarted the social media giant's ambitions. Still, with more than two billion users, Facebook may become a major player in Blockchain if it can figure out how to make it work. Salesforce's platform includes software that enables users to use Blockchain in regular business operations or to accept cryptocurrency payments.

It begins with semiconductors, as is the case with every technology. Digital currencies often use graphics processing unit (GPU) chips made by NVIDIA (NASDAQ:NVDA) and AMD (NASDAQ:AMD). Even venerable Intel (NASDAQ:INTC) has a section to collaborate with businesses creating Blockchain to promote innovation and growth. The Blockchain division of IBM (NYSE:IBM), a venerable computing company seeking to adapt to a world that is changing quickly, has worked with multiple businesses to help them use the new technology.

Blockchain services are offered on the platforms of the top three public cloud service providers, including Google Cloud (NASDAQ:GOOG), Microsoft Azure, and Amazon Web Services (NASDAQ:AMZN). In particular, Amazon might integrate these features into its vast e-commerce business. Although not a targeted bet, these cloud providers may reap significant benefits if the usage of digital ledger technology grows.

Blockchain technology, like other technologies, may provide progressive businesses a chance to develop and discover new value. Decentralized digital ledgers are now being used in financial transactions and logistics efficiency improvements, but they have a lot of potential applications in other areas of a company's operations.

If you want to wager on Blockchain's continued development, buying shares of businesses taking their time to comprehend and use the technology fully may be a solid long-term investment plan.

Investment Strategies for Blockchain Technology

Purchasing stock in any publicly traded firm that develops, uses, or invests in cryptocurrencies is a straightforward method to invest in Blockchain technology.

Equities specific to Blockchain: general businesses

The most well-known publicly traded companies that use or collaborate with Blockchain technology include:

- IBM (offers Blockchain services) (offers Blockchain services)
- Amazon (offers Amazon Managed Blockchain service)
- Intel (offers Blockchain services)
- Intel (sells GPUs to cryptocurrency miners)
- AMD (sells GPUs to cryptocurrency miners) (sells GPUs to cryptocurrency miners)
- Card: Mastercard (working on Blockchain-based cross-border payments with R3)

The Honeywell (using Blockchain to track sales)

- Use DocuSign (offers Blockchain service)
- J.P. Morgan (created its cryptocurrency, JPM Coin, and its unit for Blockchain projects)
- Israel (manufactures mining hardware)
- The Silvergate (offers banking services to Blockchain and cryptocurrency firms)

Individual Blockchain stocks: businesses involved in cryptocurrencies

As an alternative, some publicly traded businesses provide services connected to cryptocurrencies or have direct exposure to them:

- Block (offers bitcoin payment service, also holds bitcoin)
- PayPal (offers a bitcoin payment service)
- Grayscale Ethereum Trust and Grayscale Bitcoin Trust run bitcoin investment funds, respectively (operates an Ethereum investment fund)
- International Currency Exchange (operates Bakkt cryptocurrency exchange)
- Oversupply (digital retailer which accepts bitcoin)

More Bitcoin companies will become publicly listed as cryptocurrencies' usage become more widespread. For instance, the biggest cryptocurrency exchange in the United States, Coinbase, went public in 2021.

Blockchain funds

There are other options besides purchasing individual stocks to have exposure to Blockchain. According to Jones, some funds offer exposure to Blockchain technology and are primarily categorized as traditional [investment] funds.

Mutual funds and exchange-traded funds, which hold a portfolio of assets, are typically the instruments of choice for individual investors because they provide diversification and, thus, lower risk at a relatively low cost. However, they have a distinct advantage in this industry.

Bitcoin ETFs still need to come by, despite a few possibilities like the aforementioned Grayscale Bitcoin Trust: As of early 2021, the SEC in the US had declined to permit them since it was difficult to determine the value and liquidity of the currency.

These issues don't exist with Blockchain. A growing number of Blockchain ETFs are now readily available. These make investments in several businesses using Blockchain technology. These are a few of the biggest:

- Amplify Transformational Data Sharing ETF,
- Reality Shares Nasdaq NexGen Economy ETF,
- First Trust Indxx Innovative Transaction & Process ETF,
- Goldman Sachs Finance Reimagined ETF, and
- Innovation Shares NextGen Protocol ETF

Investment Advice

Despite all of its potential, Blockchain technology is still in its roots and has yet to establish itself in terms of marketable goods. According to Hadyn Jones, "Blockchain is an emerging technology just like quantum computing, electric aviation, or spatial computing, all of which need taking a risk to innovate. As a result, there are a few pointers to consider when investing in Blockchain technology.

Exercise due diligence. Nowadays, many businesses claim to be engaged in Blockchain but some invest more in the technology than others. This makes study into a specific company and its fundamentals extremely crucial. According to Jones, "the first step is to develop a case for the investment itself based on elements like the potential for development, the competitive environment, or elements that set it apart from similar ventures."

Treat the Blockchain industry as a high-risk, high-growth one: Similar to tech stocks, Blockchain equities are part of a fast-growing industry that carries significant risk for investors. It would be advisable to invest only a small amount of your available cash in Blockchain startups and diversify as much as possible in other areas because the broader utility of Blockchain still needs to be proven.

Be vigilant for new legislation and rules: researching individual companies and staying current with authorities are equally vital, especially given how much of the Blockchain industry is still in its infancy. Legislation enacted by government officials and agencies might seriously harm businesses that focus on Blockchain technology.

Understanding the regulatory environment is another helpful indicator, according to Jones. "Both the UK and EU are devoting effort in getting ready for legislation relevant to digital assets, so the underlying Blockchain technology is very much on the administrations' radar as a catalyst for growth."

Pay attention to the connection to Bitcoin: Yes, investing in Blockchain technology provides benefits over investing in bitcoin. But given that Blockchain technology has been most successfully used for bitcoin, some analysts advise focusing on businesses that primarily use it to interact with the cryptocurrency.

The COO of Digital Asset Investment Management, Adam Pokornicky,

states, "Focusing on businesses that have ties to bitcoin would be the greatest strategy to invest in businesses using Blockchain technology. Companies like Square and MicroStrategy list bitcoin as an asset on their balance sheets, or enterprises like Square, PayPal, Coinbase, Silvergate Bank, Galaxy Digital, Hive, and Voyager that build services on top of bitcoin."

Questions to Ask Before Purchasing Cryptocurrency in 2023

Investments in cryptocurrencies soared in 2020 and 2021, and many of the biggest cryptocurrencies reached record highs. Bitcoin (BTC), their grandfather, increased from about \$7,000 in January 2020 to over \$68,000 in November 2021. But things have been different in 2022.

After the Fed implemented its economic tightening measures and investors withdrew from riskier assets, prices started to decline. After that, the market experienced several shocks, each of which caused prices to drop further. One prominent instance was the Terra (LUNA) network's collapse, which had an impact on the market for several months.

Some investors believe that the worst may be behind us as we approach the new year and question if 2023 is a good year to invest in cryptocurrencies. Here are some queries to ask yourself before proceeding.

1. Do you have an emergency fund?

Before you invest in anything, whether it's stocks or cryptocurrency, make sure your emergency fund is filled. A savings account with three to six months' worth of spending will protect you from unforeseen disasters like a job loss or health problem.

The cost of cryptocurrencies plummeted in 2022, and many investors believe that prices will eventually rise again. However, you will only be able to profit from a rebound if you're forced to sell an asset when it's worth is 80% less than what you bought for it. By setting up an emergency fund, you can access it in times of need rather than liquidating investments or taking on debt.

2. Are you a long-term player?

Cryptocurrency investments, in particular, are high-risk and come with no guarantees. But if you invest with a 10- to 20-year time horizon, you can endure even sharp short-term declines like the one we've experienced this year. You must have faith in both the long-term potential of Blockchain technology and the specific projects you purchase to achieve that.

Long-term investing entails extensive study and selecting the ventures with the highest chances of success. The two biggest cryptocurrencies by market size are Bitcoin and Ethereum (ETH), so you may stick with those. As you gain experience, you might go into projects that will be useful and have a strong chance of succeeding in the ensuing decades.

3. Will you diversify your bitcoin holdings?

I'm a big fan of cryptocurrencies and hope that technology will revolutionize how we handle our online identities and spend money. However, it is now a dangerous and loosely regulated sector that faces significant challenges, and if it is unable to do so, investors risk total loss.

Only invest some of your money in cryptocurrency. Many experts advise keeping your cryptocurrency investments to 5% or less of your total capital, which is an excellent place to start. In this manner, you can make money if the industry is thriving. However, if things go right, your finances will be protected.

4. Do you have a plan?

Be sincere with yourself about your motivations for purchasing cryptocurrencies and your goals. Many cryptocurrency buyers during the 2021 crypto-frenzy did so out of fear of missing out or wanting to make quick money. Sadly, this led to consumers making purchases close to the highs needing to entirely comprehend what they were doing.

The sum of money you're willing to invest, the kinds of cryptocurrencies you intend to purchase, and the duration of your holding period should all be included in your plan. Additionally, you must understand your motivation for investing. Why do you think Blockchain technology will be successful? What factors could lead you to revise your hypothesis? That information can help you make sound decisions, which can help you avoid panic purchasing and selling.

5. Are you aware of the risks?

Investing in cryptocurrencies is incredibly dangerous. Although there is a chance for larger profits with such risks, you must know what you're entering. Cryptocurrency investing may not be for you if you lose sleep over a 20% decrease in a single day.

Here are some unsettling truths regarding investing in cryptocurrencies:

- The value of cryptocurrencies fluctuates wildly, and prices can drop sharply in weeks and might not rise to their previous highs.
- A cryptocurrency might fail on its own. You could lose everything if a cryptocurrency you possess crashes or turns out to be a hoax.
- Cryptocurrency systems and exchanges may fail. There are few consumer protections in place, so you might be unable to get your money back if the cryptocurrency exchange you use files for bankruptcy.

There are ways to limit the dangers, such as using a crypto wallet instead

of leaving your funds on a crypto market. However, not all investors would want to devote the effort it takes to comprehend how wallets operate and learn how to keep their security.

Buying Crypto in 2023

What will happen to bitcoin values in 2023 is yet to be determined. Even if more regulation improves the foundations of cryptocurrencies, it will likely generate volatility in the short term. The present crypto winter shows no indication of thawing, and prices could stay low for a long time.

If you choose to buy, don't do it hoping to make money off a rally like the one we experienced in 2021. Do it because you know what Blockchain can accomplish in the future. And even then, only invest funds you can afford to lose according to the golden rule of cryptocurrency investing.

Best Blockchain Stocks to Buy Now

1. HIVE Blockchain Technologies Ltd. (NASDAQ:HIVE) Number of hedge fund holders: 1

Leading Canadian cryptocurrency miner HIVE Blockchain Technologies Ltd. (NASDAQ:HIVE) owns and runs environmentally friendly data facilities. The business is in a good position to benefit from expanding the cryptocurrency and Blockchain sectors. Due to its affordable, secure, and dependable operations, HIVE Blockchain Technologies Ltd. (NASDAQ:HIVE) enjoys a considerable competitive edge in the Blockchain sector. One of the finest Blockchain stocks to buy right now is the stock.

On August 29, Canaccord analyst Joseph Vafi maintained a Buy rating on HIVE Blockchain Technologies Ltd. (NASDAQ:HIVE) and increased his price objective from \$5.50 to \$7. One hedge fund was optimistic about HIVE Blockchain Technologies Ltd. after the third quarter of 2022. (NASDAQ:HIVE). The fund revealed holding a \$449,000 investment in the business.

2. Marathon Digital Holdings, Inc. (NASDAQ:MARA) Number of hedge fund holders: 8

The NASDAQ-listed Marathon Digital Holdings, Inc. (NASDAQ:MARA) is a provider of enterprise-grade hosting and infrastructure for Blockchain networks and miners of digital assets. The business is one of the finest Blockchain stocks to purchase right now and is well-positioned to profit from the rising demand for Blockchain development and mining of digital assets, especially as Bitcoin and cryptocurrencies gain greater popular acceptance.

On October 7, B. Riley (analyst of Lucas Pipes) maintained a Neutral rating and reduced his price objective for Marathon Digital Holdings, Inc. (NASDAQ:MARA) from \$10 to \$8.

Marathon Digital Holdings, Inc. (NASDAQ:MARA) announced on December 6 that it mined 472 Bitcoin in November, raising its total unconstrained Bitcoin holdings from November 30 to 4,200 BTC. As of November 30, the corporation reported having \$61.7 million in cash.

Eight hedge funds reported holding long holdings in Marathon Digital Holdings, Inc. (NASDAQ:MARA) at the end of the third quarter of 2022, totaling \$9.19 million. Along with Marathon Digital Holdings, Inc.

(NASDAQ:MARA), Visa Inc. (NYSE:V), PayPal Holdings, Inc., and Mastercard Incorporated (NYSE:MA) are other stocks that are anticipated to outperform the market as the Blockchain business expands (NASDAQ:PYPL).

3. Canaan Inc. (NASDAQ:CAN) Number of hedge fund holders: 9

The leading supplier of high-performance computing systems for Blockchain applications, AI, and other cutting-edge technologies are Canaan Inc. (NASDAQ:CAN). With Application-Specific Integrated Circuit (ASIC) mining chips that are more effective and powerful than the competition, the company has a significant competitive advantage in the market. Additionally, the mining equipment it produces is highly energy efficient, which attracts more miners. Canaan Inc. (NASDAQ:CAN), one of the top Blockchain companies to purchase right now, is well-positioned to profit from the expansion of the Blockchain industry.

Canaan Inc. has a Buy rating and a \$5 price target as of July 6, according to H.C. Wainwright analyst Kevin Dede (NASDAQ:CAN). The A13 series, the company's next-generation ASIC-based bitcoin mining device, was introduced on October 24 by Canaan Inc. (NASDAQ:CAN). Model A1346 has a 110 TH/s hash rate and a 30 J/TH power efficiency, whereas the model A1366 has a 130 TH/s hash rate capacity and a 25 J/TH power efficiency.

Nine hedge funds had stakes in Canaan Inc. (NASDAQ:CAN) totaling \$8.3 million at the end of the third quarter of 2022. In contrast, eight hedge funds had positions worth \$8.9 million in the preceding quarter. With a \$4.6 million investment in Canaan Inc. (NASDAQ:CAN), Polunin Capital became the company's largest shareholder as of September 30.

4. Riot Blockchain, Inc (NASDAQ:RIOT) Number of hedge fund holders: 11

An American corporation called Riot Blockchain, Inc. (NASDAQ:RIOT) engages in three business segments: electrical products and engineering, data center hosting, and bitcoin mining. The business is well situated to profit from rising institutional and retail involvement in cryptocurrency. The company's main goal is to promote the development of the Blockchain industry through its partnerships, investments, and mining activities. One of the top Blockchain stocks to buy is Riot Blockchain, Inc. (NASDAQ:RIOT).

Unaudited production and operational updates for November were released by Riot Blockchain, Inc. (NASDAQ:RIOT) on December 5. According to reports, the company mined 521 Bitcoin in November, up 12%

from 466 Bitcoin. As of November 30, the firm could hash at a pace of 7.7 EH/s. On November 23, Riot Blockchain, Inc. (NASDAQ:RIOT) analyst Mike Colonnese began covering with a Buy rating and a \$10 price target.

Eleven hedge funds disclosed ownership of holdings in Riot Blockchain, Inc. at the end of Q3 2022. (NASDAQ:RIOT). The total value of these positions increased from \$13.63 million in the previous quarter, with thirteen positions, to \$29.36 million in the current quarter. With a position of \$8.80 million as of September 30, Balyasny Asset Management is the company's largest investor.

5. Applied Digital Corporation (NASDAQ:APLD) Number of hedge fund holders: 13

Leading next-generation data center developers and operators include Applied Digital Corporation (NASDAQ:APLD). The business offers high-performance computing solutions for digital infrastructure. Applied Digital Corporation (NASDAQ:APLD) introduced an independent fund for buying undervalued bitcoin assets on November 21. The stock is on our list of the top Blockchain stocks to purchase.

On October 12, Lake Street analyst Rob Brown maintained Applied Digital Corporation's (NASDAQ:APLD) Buy rating while lowering his price objective for the stock from \$10 to \$6.

Thirteen hedge funds owned shares in Applied Digital Corporation at the end of the third quarter of 2022. (NASDAQ:APLD). \$21.7 million was the total value of these positions, up from \$12.19 million in the previous quarter with 12 holdings. The stock is receiving favorable feedback from hedge funds.

6. Robinhood Markets, Inc. (NASDAQ:HOOD)

Number of hedge fund holders: 24

Robinhood Markets, Inc. (NASDAQ:HOOD), a leading provider of stock trading services, also provides cryptocurrency trading on its platform. The business has a sizeable market share in the cryptocurrency sector and is well-positioned to benefit from the long-term growth patterns in the global Blockchain market. One of the top Blockchain stocks is Robinhood Markets, Inc. (NASDAQ:HOOD).

On November 15, Deutsche Bank analyst Brian Bedell confirmed his Hold rating and reduced his price target for Robinhood Markets, Inc. (NASDAQ:HOOD) from \$11 to \$9. As of November 30, Robinhood Markets, Inc. (NASDAQ:HOOD) has 12.5 million monthly active users, and

the company has been expanding its user base.

Twenty-four hedge funds disclosed \$740 million worth of holdings in Robinhood Markets, Inc. (NASDAQ:HOOD) at the end of 2022's third quarter. In the prior quarter, 26 holdings totaling \$617.9 million in stakes were held. The largest stakeholder in the business as of September 30 is ARK Investment Management, which holds a \$332.39 million investment.

7. Coinbase Global, Inc. (NASDAQ:COIN) Number of hedge fund holders: 28

Coinbase Global, Inc. (NASDAQ:COIN) is a fantastic investment opportunity for investors who want to learn more about the cryptocurrency and Blockchain industries. The organization, whose goal is to provide an open financial system for the entire globe, has become a leader in the sector. As one of the most reputable and safe digital currency exchanges, Coinbase Global, Inc. (NASDAQ:COIN) is well-positioned to profit from the development of digital currencies and Blockchain technology. The best Blockchain stocks right now are ranked highly, and Coinbase Global, Inc. (NASDAQ:COIN) is one of them.

On November 22, Needham analyst John Todaro maintained a Buy rating on Coinbase Global, Inc. (NASDAQ:COIN) while lowering his price objective from \$89 to \$73. Twenty-eight investors had \$976.8 million shares of Coinbase Global, Inc. (NASDAQ:COIN) in their portfolios at the end of 2022's third quarter. ARK Investment Management revealed a position in the company worth \$497.6 million, making it the largest investor.

8. International Business Machines Corporation (NYSE:IBM) Number of hedge fund holders: 40

International Business Machines Corporation (NYSE:IBM) has been making progress in the Blockchain industry. The business has become a market leader in enterprise Blockchain services and is ideally situated to gain from the sector's positive trends. Organizations may safely store, manage, and share data with third parties using the company's distributed ledger technology (DLT) platform. It is built on the open-source Hyperledger Fabric platform, developed in a cooperative effort to enhance Blockchain technologies across several industries. International Business Machines Corporation (NYSE:IBM) comes in sixth place regarding the top Blockchain stocks.

International Business Machines Corporation has positive sentiment on Wall Street (NYSE:IBM). In October, Morgan Stanley analyst Erik

Woodring restated his Overweight rating on shares of International Business Machines Corporation (NYSE:IBM) and lowered his price objective from \$155 to \$152. On October 17, International Business Machines Corporation (NYSE:IBM) analyst David Grossman raised his price objective for the stock from \$150 to \$140 while keeping a Buy rating.

Forty investors revealed holdings in International Business Machines Corporation (NYSE:IBM) with a total value of \$868.7 million at the end of the third quarter of 2022. The largest shareholder in the business as of September 30 was Arrowstreet Capital, with a \$515.76 million holding.

International Business Machines Corporation (NYSE:IBM), Mastercard Incorporated (NYSE:MA), Visa Inc. (NYSE:V), and PayPal Holdings, Inc. are a few of the businesses anticipated to continue to lead the global Blockchain sector for years to come (NASDAQ:PYPL).

Key Features in this Chapter

- One of the top Blockchain stocks to buy is Riot Blockchain, Inc. (NASDAQ:RIOT).
- 24 hedge funds disclosed \$740 million worth of holdings in Robinhood Markets, Inc. (NASDAQ:HOOD) at the end of 2022's third quarter.
- 28 investors had \$976.8 million shares of Coinbase Global, Inc. (NASDAQ:COIN) in their portfolios at the end of 2022's third quarter.
- The business has become a market leader in enterprise Blockchain services and is ideally situated to gain from the sector's positive trends.
- International Business Machines Corporation (NYSE:IBM) comes in sixth place regarding the top Blockchain stocks.

BLOCKCHAIN-AS-A-SERVICE (BAAS)

Blockchain-as-a-Service (BaaS): what is it?

The term "Blockchain-as-a-service" (or "BaaS") refers to a cloud-managed platform that enables individuals and companies to create rapidly, host, and maintain Blockchain applications. The acronym "BaaS" is also frequently used to refer to "backend as a service."

The establishment and administration of cloud-based networks by a third party for businesses that develop Blockchain applications are known as Blockchain-as-a-service (BaaS). These third-party services are a recent innovation in the rapidly expanding field of Blockchain technology. The usage of Blockchain technology has expanded to include secure transactions of all kinds, going far beyond its most renowned use in bitcoin transactions.

With Blockchain-as-a-service, a third party manages the infrastructure and backend, decreasing the barrier to adoption for people and businesses. There is a need for hosting services as a result.

Learning about Blockchain-as-a-service (BaaS)

BaaS operates similarly to the software as a service (SaaS) concept and is, in fact, built on it. Using cloud-based solutions, customers can create, host, and manage their Blockchain apps and related Blockchain services. The cloud-based service provider keeps the infrastructure adaptable and functional simultaneously. BaaS is considered a breakthrough in the larger Blockchain ecosystem that will increase the business use of Blockchain.

Leading companies in the BaaS market include:

- In 2015, Ethereum Blockchain-as-a-service was made available on Microsoft Azure thanks to a partnership between Microsoft and ConsenSys.
- Amazon, which unveiled Amazon Managed Blockchain, a solution that "makes it easier to construct and administer large Blockchain networks" using open-source frameworks like Hyperledger Fabric and Ethereum.

- R3, a collaboration of international financial institutions, created Corda, a distributed financial ledger.
- PayStand, which focuses on transferring money between businesses.

Businesses and consumers alike are becoming more open to adopting Blockchain technology. However, the technical challenges and administrative burden associated with developing, establishing, running, and maintaining a Blockchain's infrastructure frequently serve as a barrier.

For a price, BaaS provides an outside service provider to establish all required Blockchain infrastructure and technologies. After it is constructed, the provider keeps taking care of the client's complicated backend processes. The BaaS provider often provides support services, including bandwidth management, appropriate resource allocation, hosting needs, and data security features. The BaaS provider frees the client to concentrate on the Blockchain's fundamental functions.

An illustration of Blockchain-as-a-service (BaaS)

The Hyperledger project's Blockchain-as-a-Service Hyperledger Cello, a BaaS-like Blockchain module toolkit and utility system. The function of a BaaS provider is comparable to that of a web hosting company. The website's content is created and maintained by its developers on their personal computers. They might engage support personnel or sign up with a third-party host like HostGator or Amazon Web Services. These outside companies handle the infrastructure and upkeep difficulties.

BaaS might be the impetus for a larger and deeper use of Blockchain technology across numerous business sectors and industry verticals. No matter how big or small, an organization may outsource the technically challenging job and concentrate on its core competencies rather than building and managing its own Blockchains.

How Does BaaS Work?

Blockchain is a requisite component of this budding technology. With the aid of Blockchain technology, organizations can communicate data quickly and securely because the transaction is not controlled by any one party.

Building, maintaining, and monitoring a Blockchain system for a particular application takes too much time because it is typically advanced. Distributed ledgers are a more widely used alternative to Blockchain-as-a-service (BaaS), especially for organizations finding ways to reduce overhead costs. Although Blockchain technology has many sensible uses, some sectors still need help realizing their full potential.

How can businesses benefit from Blockchain?

Through cloud computing, BaaS (Blockchain-as-a-Service) enables businesses to build and maintain Blockchain-based networks. Offering third-party services is a relatively recent development in Blockchain technology. Blockchain applications have developed to include safe transactions of various kinds, going far beyond its most well-known use in bitcoin transactions. Hosting services are, therefore, highly sought after.

Through the cloud-as-a-service concept, Blockchain technology is now being used to accommodate services for a larger audience, similar to how SaaS, PaaS, and IaaS models enable the utilization of cloud-based apps and storage. Businesses can use BaaS as a low-cost way to effectuate Blockchain technology. BaaS makes it possible for businesses of any size to make Blockchain apps using the services of Blockchain providers.

Fortune Business Insights says that the global BaaS market will grow at a CAGR of 39.5% between 2020 and 2027, taking it to USD 24.94 billion by 2027.

How important is Blockchain-as-a-Service?

Information is the engine that keeps a company forward. It is recommended to get it delivered as promptly and accurately as feasible. Blockchain is the best way to deliver that information because it offers real-time, shareable, and transparent data that is kept on an unchangeable ledger and is only accessible to people who have permission to access it. Orders, payments, accounts, production, and much of the above are tracked via Blockchain networks. Members can view all the information about a transaction in one spot, increasing their trust in creating more efficiency and opportunities.

IT companies in various industries are actively researching the implementation of Blockchain technology. Despite this, plans for adapting are frequently slowed down by the intrinsic technical complexity, operational overhead costs of constructing, managing, and maintaining the Blockchain, and a lack of subject expertise. However, BaaS is currently considered a likely remedy for this issue.

By providing businesses with access to qualified Blockchain developers, process and governance experts, and the necessary cloud infrastructure, the right Blockchain-as-a-service provider can make it easier for them to transition to Blockchain technology without worrying about startup and overhead costs.

A reliable BaaS supplier will also give access to an abundance of knowledge and experience that can be used to upgrade system security. As a result, it drastically lowers the risks that would be managed if it had been created internally.

The function of Blockchain-as-a-Service

When a third party puts up all the necessary "Blockchain technology and infrastructure" for you, it is known as BaaS. You pay the provider to set up and maintain the nodes. For users and their businesses, BaaS providers manage the backend dynamic.

The BaaS operator ensures that mandatory Blockchain-related assets and services are carefully preserved and managed. It can also control bandwidth, determine storage requirements, distribute resources, and pinpoint security threats. BaaS relieves users from the worry of infrastructure and performance issues so they can focus on their primary activity, Blockchain technology.

Take BaaS as an illustration for a hosting company. Create a web page that millions of people see daily in only a few minutes. You may manage your website from the comfort of your office using your computer or server, and you can either handle the work yourself or contract support staff. Additionally, a website could be hosted in a place that is inconvenient for you.

Hyperledger Cello, a utility system and toolkit for Blockchain modules that resemble a BaaS platform is a nice illustration of Blockchain technology. A distributed computing architecture called Hyperledger Cello (HLC) enables more effective management and use of Blockchain systems.

Blockchain Cloud Comparison

In this section, we examine the identical cloud services offered by several cloud providers. We will look at the parallels, contrasts, and any other relevant information. We'll then delve into the Blockchain options that cloud providers have to offer in this piece.

What applications use Blockchain?

Blockchain technology is now used for various purposes, including supply chain traceability, retail loyalty incentive schemes, real estate transactions, and electronic voting.

Google Cloud Blockchain services

Google Cloud Platform (GCP) does not currently provide any Blockchain services. You could download and install a Blockchain platform on a few virtual machines, but you'd still need to manage everything.

Microsoft Azure Blockchain services

By terminating its Azure Blockchain Service offering in September 2021, Microsoft Azure left the Blockchain-as-a-service industry. The Quorum Blockchain-as-a-service on the Azure Cloud is offered by ConsenSys, a third party, to customers of that service. However, this does not imply that Microsoft Azure is giving up on Blockchain altogether.

Microsoft has included Blockchain technology in a few brand-new products now under preview. Blockchain technology is used to power a new ledger feature for the Azure SQL Database that improves record integrity and lets users demonstrate that records haven't been altered.

For the same reason, Microsoft has also released Azure Confidential Ledger. Your data is kept on your Azure storage account as objects instead of in a SQL database, except Confidential Ledger.

AWS Blockchain services

The complete option for anyone wishing to board the Blockchain train is Amazon Web Services (AWS). AWS offers Managed Blockchain if you search for a fully automated Blockchain solution. Customers can start a node on the open Ethereum Blockchain network by pressing a few buttons.

Users can quickly set up or join private Hyperledger Fabric networks with a few clicks. You don't need to worry about technicalities because these public nodes and private networks are completely managed; you pay by the hour.

Amazon Quantum Ledger Database (also available through AWS) is designed for programmers who guarantee completely immutable data change tracking. The shortened name for this is QLDB, and it has two components: (1) a section that keeps the current state data and (2) an append-only journal that records every alteration made to that current state database. Developers and applications can communicate with it using SQL or the API since it is essentially simply a database. Behind the scenes, the Blockchain ledger portion is managed.

If you're familiar with other databases, you'll notice that this ledger sounds like transaction logs or journaling. And the reason is that it is something similar. The distinction is that every entry in that journal is cryptographically linked to both the entry before it and the one after it. This makes it simple to verify and unchangeable.

Some well-known names have signed on to use AWS's Blockchain offerings, including The BMW Group, Nestle, Sony Music, and Workday are a few of the well-known companies that have used AWS's Blockchain solutions.

Additionally, AWS claims to host at least 25% of all Ethereum network nodes and has over 70 validated Blockchain solutions available on its AWS Marketplace. But is AWS the only provider of Blockchain-as-a-service? In no way.

IBM Cloud

IBM Cloud was one of the first prominent cloud providers to enter the hosted Blockchain market. It has one of the most developed offerings as a result. IBM hosts Blockchains for several major firms, such as Walmart and the shipping company Maersk. Additionally, IBM has significantly played a

part in Hyperledger Fabric, a renowned open-source Blockchain technology, and the software that powers its branded Blockchain products, and it continues to do so.

Oracle Cloud

Then there's the Oracle Blockchain Platform which is a Hyperledger Fabric implementation with additional capabilities added to improve compatibility with the Oracle Cloud. They host Blockchains for the financial services, fashion, healthcare, and transportation sectors. Blockchain is merely another tool in their customers' arsenal, and both Oracle and IBM use their partnerships with large enterprises to market their Blockchain services.

Other Blockchain services in the cloud

Numerous cloud-native firms outside the major cloud providers are working to capitalize on Blockchain through industry-specific solutions or other distinctive value propositions. Just think carefully about who you might connect with because Blockchain use cases are typically rather far-reaching. Thus, the existing dispersed and diverse cloud landscape for Blockchain is described. Others are remaining steadfast and increasing their investments in Blockchain-as-a-service, while public clouds like Azure and SAP Cloud may have backed away.

Blockchain-as-a-service is currently trudging through what Gartner calls "the trough of disillusionment," according to the renowned Hype Cycle from Gartner. This means that no pun intended, all those high expectations are being discussed.

The following key aspects of Cello are supported by Blockchain technology and contemporary PaaS tools:

- Automate the Blockchain network lifecycle's creation, termination, deletion, and maintenance.
- Permitting individualized network types and sizes for Blockchain networks.
- Support native Docker hosts, virtual machines, vSphere, bare-metal,

Swarm, and Kubernetes

- The platform delivers cutting-edge features like monitoring, logging, health, and analytics capabilities by integrating current tools like ElasticStack.

Blockchain-as-a-service might serve as the impetus for the widespread adoption of Blockchain across numerous businesses and industries. To focus on their core businesses rather than managing and creating their Blockchains, big or small companies can now easily "outsource" their technologically arduous duties.

For their own BaaS arrangements, many IT corporations have already thrown their hats into the Blockchain ring. The Azure platform from Microsoft is available to support the BaaS system. While Amazon and Oracle provide cloud services, their own BaaS focuses on private Blockchains.

The bottom line

Big tech firms are already providing Blockchain to help customers adopt this revolutionary technology and maintain competitiveness. Large companies like Microsoft and IBM deliver Blockchain-as-a-service. Instead of creating these platforms themselves, CEOs can source Blockchain functionality thanks to their services. Since Blockchain technology has so much promise, a Blockchain-based service might be a game-changing breakthrough in any business.

If you want to advance your business, BaaS is the technology you need. Blockchain-as-a-service expands the potential of your business by making Blockchain technology available to a larger audience and supporting all the quickly developing use cases of the technology.

Key Features in this Chapter

- The establishment and administration of cloud-based networks by a third party for businesses that develop Blockchain applications are known as Blockchain-as-a-service (BaaS).
- We examine the similar cloud services offered by several cloud providers.
- Fortune Business Insights says that the global BaaS market will grow at a CAGR of 39.5% between 2020 and 2027, taking it to USD 24.94 billion by 2027.
- Blockchain-as-a-service might serve as the impetus for the widespread adoption of Blockchain across numerous businesses and industries.

BLOCKCHAIN AND WEB 3.0

Web 3.0 explained

Public Blockchains, a database renowned for facilitating bitcoin exchanges, are the foundation of Web 3.0, a potential future version of the internet. Web 3.0 is intriguing because it is decentralized, which implies that users own and manage some areas of the internet instead of customers using services that companies like Google, Apple, or Facebook mediate.

Due to the lack of "permission" or "trust" in Web 3.0, centralized authorities cannot regulate who has access to what services. Virtual transactions between two or more parties can also happen without needing a go-between. Because these companies and intermediaries do the majority of data collecting, Web 3.0 protects user privacy better.

Decentralized finance, also called DeFi, is a Web 3.0 feature fast gaining popularity. Real-world financial transactions must be fulfilled on the Blockchain without the assistance of banks or the government. In the meantime, many large venture capital firms invest heavily in Web 3.0, and it isn't easy to imagine that their involvement won't lead to some centralized power.

In this chapter, we discuss how the web has changed through time, why Web 3.0 is a hot topic, what it's used for, what Web 3.0 in crypto means, where it's going next, and why it matters.

The Evolution of the Web

The primary instrument used by billions of people to read, write, and exchange information, as well as to contact others online, is the World Wide Web. The Web has undergone significant transformation over the years, and the apps it uses today are almost recognizable from those of its infancy. The web has undergone three stages: Web 1.0, Web 2.0, and Web 3.0.

What is Web 1.0?

Web 1.0 refers to the initial iteration of the internet. Think of the read-only or syntactic web as Web 1.0. The majority of participants were consumers of content. In contrast, most makers were web developers who created websites with content generally given in text or visual format. In general, Web 1.0 was around from 1991 until 2004.

In Web 1.0, static content was distributed via websites instead of dynamic HTML. A static file system was used instead of a database for data and content, and users could only do a little on the websites.

What is Web 2.0?

Most of us are only familiar with the web's current iteration, commonly referred to as Web 2.0, the interactive read-write and social web. In Web 2.0, you don't have to be a developer to participate in the development process. Many apps have been created so that anyone can make them.

You can think and express your thought to the world. In Web 2.0, you may also submit a video and make it accessible to millions of other users for viewing, participation, and commentary. Web 2.0 applications include YouTube, Facebook, Flickr, Instagram, Twitter, and other social media.

Companies may create new concepts that let people contribute more to the Social Web thanks to web technologies like HTML5, CSS3, and Javascript frameworks like ReactJs, AngularJs, VueJs, and others. Because Web 2.0 is centered on users, developers merely need to create a system to enable and engage users.

Think about how Instagram, Twitter, LinkedIn, and YouTube, four

popular apps, were in their early years compared to how they are now. All of these businesses frequently follow the following process:

- The business releases an app.
- It accepts the greatest number of students.
- After that, it monetizes its user base.

The user experience is frequently slick when a developer or company develops a well-known app, especially as the app's popularity increases. This is what has allowed them to gain traction so quickly initially. Many software companies initially give monetization little thought. Instead, they are only concerned with attracting and keeping new customers, yet they eventually need to start making money.

However, the limitations of accepting venture money frequently affect the lifespan and, in the end, the user experience of many of the services we currently use. For instance, investors often expect a return on investment in the tens or hundreds of times what they invested when a company receives venture financing to develop an application. This indicates that the organization is frequently steered along one of two paths: marketing or data sales, rather than following a long-term growth strategy that can be sustained organically.

For various Web 2.0 businesses like Google, Facebook, Twitter, and others, more data translates into more targeted advertisements. More clicks are generated as a result, which increases ad revenue. Utilizing and consolidating user data is essential for the web to work as we currently know and use it. Data breaches consequently frequently occur in Web 2.0 apps. Even websites that detect data breaches and notify you when your personal information has been compromised exist.

In Web 2.0, you cannot control how your data is saved. In reality, companies regularly collect and store user data without their consent. All this data is then owned and managed by the companies in charge of these platforms. Furthermore, governments frequently shut down servers or seize bank accounts when they suspect someone is expressing an opinion that is in opposition to their propaganda. Governments can use centralized servers to interfere with, manage, or stop certain applications.

Since banks are centralized and digital, governments frequently intervene in them. They may, however, freeze bank accounts or restrict access to funds

during extreme volatility, excessive inflation, or other political unrest. Web 3.0, which aims to rethink how we create and use applications fundamentally, will address many of these issues.

What is Web 3.0?

The Web 3.0 era, also called the Semantic Web or read-write-execute, started in 2010 and shows where the internet is going in the future. Artificial intelligence (AI) and machine learning (ML) make it possible for computers to look at data like humans do. This lets computers create and share helpful information tailored to users' needs.

Web 2.0 and Web 3.0 are based on decentralization, but there are some major differences. Developers of Web 3.0 applications rarely make and sell ones that store data on just one server or database that is usually hosted on and managed by a single cloud provider.

On the other hand, Web 3.0 applications are built on Blockchains, decentralized networks of many P2P nodes (servers), or a combination of both. The Web 3.0 community often calls these programs "DApps," which stands for "decentralized apps." For a decentralized network to be reliable and safe, users (developers) must be paid for providing the best services.

What does Web 3.0 in crypto mean?

You'll see that cryptocurrency is regularly brought up when discussing Web 3.0. This is because cryptocurrencies are a big part of many Web 3.0 protocols. Instead, it gives money (in the form of "tokens") to anyone who wants to help build, run, contribute to, or improve one of the projects. The goal of creating a decentralized internet is linked to digital assets called "Web 3.0 tokens." These protocols could provide various services such as hosting, computing, bandwidth, storage, identity, and other internet services that cloud providers used to offer.

For example, people who make video infrastructure and streaming services can sell their products on the Livepeer protocol based on Ethereum. In the same way, Helium uses Blockchains and tokens to get individuals and small businesses to provide and test wireless coverage and send device data across the network.

The protocol gives people a wide range of technical and non-technical ways to make a living. Users of the service often pay to use the protocol, just like they would pay a cloud provider like Amazon Web Services. Decentralization usually means eliminating intermediaries who aren't needed and often waste time and money.

Web 3.0 will also have important parts like non-fungible tokens (NFTs), digital currencies, and other parts of the Blockchain. For example, Reddit wants to get into the Web 3.0 space by making a system that lets users manage aspects of the online communities they are a part of using cryptocurrency tokens. According to the idea, users would use "community points," which they would get by posting on a certain subreddit. How many people vote up or down on a post determines how many points that user gets. (On the Blockchain, it's just Reddit Karma.)

These points can be utilized as voting shares, giving users who have contributed significantly more influence over decisions that have a more considerable impact on the community. These points can't just be taken away because they are kept on the Blockchain and follow you, giving their owners more control. Fair enough, this is simply one use of a Web 3.0 concept called Decentralized Autonomous Organizations (DAOs), which employs tokens to share ownership and decision-making power more fairly.

Web 2.0 vs. Web 3.0

What characteristics does Web 3.0 have?

The change from Web 2.0 to 3.0 is happening slowly, but only a few people know. Web 3.0 applications look like Web 2.0 applications, but their back ends differ. In the future of Web 3.0, there will be universal applications that can be accessed and used by different hardware and software. Together, we'll have more time for both work and play.

Emerging technologies like distributed ledgers and Blockchain storage will enable data decentralization and the construction of a transparent and secure environment in contrast to the centralization, monitoring, and exploitative advertising of Web 2.0.

When decentralized infrastructure and application platforms replace centralized tech firms, people can legitimately own their data in a decentralized web. To further understand the nuances and intricacies of Web 3.0, let's examine its four key characteristics.

Semantic Web

The "Semantic Web" is an essential part of Web 3.0. Tim Berners-Lee talked about a network of information that computers can analyze. So, what does that mean in everyday language? What does it mean to talk about "semantics"? What's the difference between "I love Bitcoin" and "I <3 Bitcoin"?

Even though the two phrases are put together differently, they mean the same thing. Semantics is the study of what facts mean or how they make us feel, and in the example above, both phrases make us feel the same way. Artificial intelligence and the semantic web are Web 3.0's two pillars. By helping the computer learn what the data means, the semantic web will enable AI to create practical use cases that will better use the data.

The main idea is to create a knowledge spiderweb on the internet that will help with word meaning comprehension and content creation, sharing, and connection through search and analysis. Semantic information in Web 3.0 will make data transmission easier. The user experience advances due to a new degree of connectedness that uses all available data.

3D graphics

The future of the internet will change as Web 3.0 transforms it from a straightforward two-dimensional Web to a more realistic three-dimensional cyber world. Three-dimensional design is widely used in Web 3.0 websites and services, including e-commerce, online games, and the real estate market.

While it may seem unusual to think that thousands of individuals are currently communicating here from around the world, it is true. Think of online games like World of Warcraft or Second Life, where players are much more concerned with the welfare of their virtual avatars than their actual counterparts.

Artificial Intelligence (AI)

Artificial intelligence will enable websites to screen and present consumers with the most relevant information. In the present Web 2.0 era, businesses have asked for consumer feedback to learn how well a good or asset works. Think about a website like Rotten Tomatoes, where people may rate and review movies, as an illustration. Higher-graded movies are frequently thought of as "good movies." This kind of list enables us to quickly move from the "bad data" to the "excellent data."

As we've already said, peer reviews are among Web 2.0's most important contributions. However, as we all know, human recommendations are not impervious to bias. A group can work together to give a movie inflated positive reviews to boost its ratings. Artificial Intelligence can differentiate between reliable and unreliable data and provide accurate information.

Ubiquitous

Ubiquitous is a term used to describe the idea of existing or being present in several places at once, also known as omnipresence. Web 2.0 already includes this feature. Consider social networking sites like Instagram, where users may upload and share the photographs they shoot with their phones to make them their intellectual property. The image becomes widely accessible after being placed online.

The Web 3.0 experience will be available anywhere, at any time, thanks to the development of mobile devices and an internet connection. Compared to Web 1.0 and Web 2.0, your desktop computer and smartphone will no longer be the only devices you can use to access the internet. It will have infinite power. Web 3.0 is sometimes referred to as the web of everything and everywhere because most objects around you are connected online (Internet of Things).

Blockchain Technology and Web 3.0 are the Upcoming Big Things

Great minds can often change the world with just one invention. We would still be sitting around a candle or writing a letter to get in touch with a friend instead of just unlocking our phones, logging into Facebook, and messaging friends. Or, we would still be sending pigeons on errands, just like the Greeks did 3,000 years ago when they used pigeons to let the rest of the world know who won the Olympics.

It is evident that anything a scientist discovers or invents is based on earlier discoveries and inventions, according to Lukas Foss. The same is true for Web 3.0 and Blockchain. We humans have been actively shaping the future based on what we believe we need and how much more effectively we can act.

The role of Blockchain in Web 3.0

One of the most significant technological developments since the internet is a game-changing innovation that will change how the internet operates. Due to Blockchain technology, every industry is expected to experience significant upheaval in the coming years—a network where clients can agree without needing to believe in one another. However, our understanding of Blockchain needs to be improved to fully comprehend its concepts.

Technology has, until now, been the purview of computer programmers, authorities, and world leaders. The internet was a centralized authority system when it first appeared, but the Blockchain is different. With the help of Blockchain technology, the global tech ecosystem will become more democratic, giving each user control over their data because the infrastructure for computing and storage will be owned and run by the community. The government or business entities would no longer have control over this.

The simplest way to understand Blockchain is probably to conceive it as returning to the original digital ideal of pure democracy while incorporating security mechanisms that weaken the dynamics that encourage the consolidation of gatekeepers. In a nutshell, the architecture fulfills the initial goal.

Web 3.0

Web 3.0 is currently without an official definition, but it is the next step in developing websites and apps on the internet. This web will be supported by AI, Blockchain, IoT, 5G, Metaverse, NFTs, and cryptocurrencies. As the new web version eliminates the need for a central authority to oversee the network in favor of a democratically governed distributed consensus, it paves the way for a decentralized economic system.

Decentralized apps (DApps), which run on peer-to-peer networks and use tokens to reward users, will be a part of Web 3.0. They are free to use and may be enhanced by anybody in the community. This area is still under development. Even though new DApps are created daily, they all have these key traits.

1. Open source
2. Motivated (tokens)
3. A continuous Blockchain
4. Decentralized
5. Use smart contracts

Decentralizing the internet would involve replacing centralized data hosting services with a peer-to-peer architecture. Due to incidents involving extensive espionage and content restriction, there needs to be more trust in network maintenance groups, which has led to an interest in the decentralized web.

The future

Blockchain must grow before it can be used everywhere, and businesses need to start planning and thinking about their Blockchain strategy immediately. Blockchain technologies might be considered a tool to introduce the next level: sharing value, in the same way, that the internet is a means of sharing information.

We are now experimenting with Blockchain technology, much like the early days of the internet, hoping it will revolutionize how the world works. It will take time for the web to decentralize, and we must bring the decentralized internet together while allowing our perspectives to change.

This decade should be reframed as visionaries and change-makers because of the incredible amount of change that occurred over the past 30 years and could potentially happen over the next 20 years.

One of the fascinating aspects about Web 3.0 and Blockchain is that I fully expect this piece of information to be obsolete or weak in a statement, owing to how much more new material will be released in a short period, which is why I'm so excited about it.

In conclusion, the trouble with truly brilliant innovations like this is that we can never predict what will happen to them. We can only fully comprehend their development and impact on us in retrospect, many years later. Foresight is tough; looking back is simple. Let's examine a few of the earlier discoveries that transformed our world:

1. The printing press in the 1400s; Before the printing press's creation, sound education and information were only accessible to the affluent. When the printing press was created, the gap in learning and information was entirely closed.
2. The steam engine arrived and altered the way we operate. Enslaved people performed most of the labor and production, but the steam engine's development significantly reduced slavery and colonialism.
3. The internet, created sixty years ago, has since changed the information industry and how people communicate. In fact, it has made the world significantly smaller than it was sixty years ago.

Blockchain and Web 3.0 are like the printing press and steam engines in the early modern era. Lastly, let's imagine a case where:

- You manage your financial resources, not banks.
- You can send money instantly and at virtually no cost to anyone, anywhere in the globe.
- When electoral results are viewed on a publicly immutable ledger, Blockchain technology will be used for voter registration, identity verification, and electronic vote counting to guarantee that only legitimate votes are counted and that none are changed or removed, making elections fairer and more democratic.
- Because it is available and verifiable on the system, Blockchain can

help to simplify and increase the security of the distribution of welfare, unemployment payments, etc., to the general public.

- A future in which medical records can be safely stored on the Blockchain by healthcare facilities and shared with only patients or authorized doctors. This will enhance data security and may potentially speed up and increase the precision of diagnosis.

Surprise! We already exist on that planet, and Blockchain and Web 3.0 are already achieving it. These technologies have countless potential applications. The direction we take from here depends entirely on our collective imaginations because here is the future.

Best Web 3.0 Coins to Invest for the Long Term in 2023

Records were broken in the cryptocurrency sector during the bull market of 2020–21. Even though 2022 has been a testing year, now is a great investment opportunity due to the recent decline in cryptocurrency pricing. This section can help you if you're looking for the top Web 3.0 coins for 2023 and beyond. You can find out which Web 3.0 coins are the greatest long-term bets in 2023.

1. Metacade – best overall

The first virtual arcade on Web 3.0 is Metacade. It is an all-in-one platform that offers simple access to a variety of the greatest GameFi and P2E games. Gamers can access the Metacade to play competitively against the world's top players, socialize with other players, and play casually with their friends. Additionally, Metacade creates a plethora of fresh chances for Web 3.0 passive income. You can make money on Metacade, for instance, by:

- Participating in gaming competitions and taking home prizes
- Publishing interesting stuff
- Prize draws for winners
- MCADE token staking
- and more

The Metacade team has ambitions to run a job portal where Web 3.0 businesses can meet outstanding candidates. As chosen by the community, Metacade's funding program, Metagrants, will give developers the assistance they need to create games for the platform. These components comprise one of Web 3.0's most ambitious and exciting new projects. The Metacade genuinely has something that everyone can appreciate. Additionally, the project is still holding its presale, allowing you to join before publicity campaigns begin.

You can purchase 125 \$MCADE at the beginning of the presale for one dollar. This price will rise to 50 \$MCADE for \$1 at the end of the presale. It is best to become involved as soon as possible.

2. Chainlink – Oracle makes Web 3.0 innovation possible

An infrastructure token called Chainlink frequently needs to be noticed. However, it can be a wise investment in the future. If you need to become more familiar, Chainlink is a price oracle. To track the results of real-world events, update on-chain tokens that track stock values, and perform other tasks, Blockchains and apps use it to obtain real-time data.

Aave, Trader Joe, and other leading cryptocurrency projects like these already use Chainlink. Each project must buy and hold substantial Chainlink tokens to obtain the required information, and Chainlink's cost will rise as it grows more popular.

3. Avalanche: a different layer-1 with a special upside

Avalanche is an Ethereum rival with a distinctive scaling strategy. Instead of layer-2s, it scales using a system of subnets. These advantages enable projects to build a Blockchain that satisfies technological requirements without producing security. Additionally, Avalanche is quickly becoming one of the leading companies for Web 3.0 game initiatives. AVAX's value might soar if that emerges as one of the dominant narratives of the upcoming bull market.

4. Treasure (MAGIC) – the leading GameFi project by Arbitrum

On Arbitrum, Treasure is a hub for communication between NFTs, GameFi, and DeFi. By sales volume, it presently has the largest NFT market on Arbitrum and already has many high-quality titles available. Treasure is one of the best projects on the Ethereum layer-2 called Arbitrum, quickly becoming one of the best. If Arbitrum keeps expanding, Treasure ought to grow significantly as well.

5. DeFi Kingdoms: a P2E RPG with potential for a multichain

During the previous bull market, DeFi Kingdoms was one of the most discussed GameFi projects on Crypto Twitter. In this point-and-click role-playing game, you may buy heroes with various NFTs and level them up through quests and combat, all while generating passive revenue.

The project has had a challenging 2022, and the team working on it is challenging to release updates at the promised pace. DeFi Kingdoms, however, has a lot going for it, and if the team can resume its superior performance throughout the following bull market, this undertaking could do remarkably well.

6. Uniswap – the premier decentralized exchange

A project like Uniswap hardly requires an introduction. It's the first significant decentralized trading protocol and one of the most popular uses of

cryptocurrencies. The issue with Uniswap has always been that its token only truly gains value by using the network or from charging users. This may change soon, and the token may begin to amass some of the value the platform continues to generate. In that case, the next bull run might see Uniswap's price rise dramatically. But even if it doesn't, Uniswap has made a name for itself as the top platform for decentralized trading, which should help it thrive in the upcoming cycle.

7. Solana - a different layer-1 with significant benefits

Like Avalanche, Solana is an Ethereum rival that stands out due to its unique scaling strategy. Because Solana is naturally scalable, it can handle more transactions per second without using layer-2s, subnets, or any other additional infrastructure.

Solana is now one of the most consumer-friendly smart contract platforms available as a result. Additionally, it boasts a vibrant NFT community, and this may help it keep the market share it has established for itself even as new rivals keep entering the market.

8. The future of online identification: Ethereum Name Service (ENS)

Another token to consider is Ethereum Name Service (ENS). The most popular smart contract platform today is Ethereum, and its address system is ENS. This makes it simpler to communicate goods to friends and identify oneself in the Web 3.0 environment.

If you type Bob.Eth instead of a long list of random numbers and letters for their address, you could use your ENS name to send a friend \$50 for dinner. If Ethereum maintains its dominance in this market, ENS names may appear more frequently. And when that occurs, the ENS token's value should increase significantly.

9. Web 3.0 real estate and virtual world - Decentraland

One of the largest metaverse projects during the previous crypto bull market was Decentraland, which may still have plenty of room to develop. The initiative enables users to purchase, sell, and explore virtual land that they may even construct. For instance, some land plots have casinos, and the owners receive money from players who gamble in-game.

The ability for players to personalize their avatars with various in-game wearables allows designers to make these wearables and sell them on the Decentraland servers to make money. The fact that there is much to be done in the Decentraland community should help the token price of the project in the future. This is particularly true as the game keeps becoming better and

more polished.

10. Filecoin – decentralized data storage

Filecoin is a different project you should look at if you advocate decentralization and Web 3. It is a platform for decentralized file storage that enables users to save documents, audio files, and movies independently of a centralized service provider like Google. The benefit is that any content, regardless of its type, can be stored indefinitely. Additionally, users of Filecoin can be reassured about their data being misused for commercial gain or disclosed without permission.

Filecoin should gain popularity as more and more people continue to value privacy. This will be wonderful news for owners of FIL, especially those who start investing in it before the general public.

11. Binance Coin (BNB) - a major ecosystem's currency

Another initiative that's worthwhile looking into is Binance Coin. The most distinguished cryptocurrency exchange in the world is Binance. Additionally, it developed and ran the Ethereum-rivaling Binance Smart Chain smart contract platform. The platform's native currency is BNB.

Because so many people utilize Binance Smart Chain and Binance as an exchange, BNB might be a worthwhile investment. Binance Smart Chain was the preferred platform for launching new projects during the previous bull run because doing it on Ethereum was too expensive.

Investors might observe similar events throughout the upcoming bull run. As a result, more users will join the Binance Smart Chain ecosystem; and to interact with the chain, they will all need to purchase BNB. The market cap of Binance Coin should significantly increase as a result of that operation.

Metacade is superior to the others

Each currency on this list is one of the greatest Web 3.0 coins to invest in for the long term in 2023 and beyond. The MCADE token from Metacade would be the first choice for a reason, though, if you had to pick just one. The best Web 3.0 gaming, money-making, and community-building options are available from a single platform, thanks to Metacade. From competitive gaming tournaments and prize draws to passive revenue through staking and content uploading, it has something to offer everyone.

Whatever your Web 3.0 objectives or who you are, Metacade will help you achieve them. This enormous potential distinguishes Metacade as the top

Web 3.0 currency to hold for the long term in 2023.

Key Features in this Chapter

- Public Blockchains, a database best known for facilitating Bitcoin exchanges, are the foundation of Web 3.0, a potential future version of the internet.
- In this chapter, we discussed how the Web has changed through time, why Web 3.0 is a hot topic, what it's used for, what Web 3.0 in crypto means, where it's going next, and why it matters.
- In Web 2.0, you don't have to be a developer to participate in the development process.
- Web 2.0 is centered on users, and developers merely need to create a system to enable and engage users.

PREDICTIONS FOR 2023: WHAT TO EXPECT FROM AI, BNPL, CRYPTO, AND MORE

Our researchers examine key trends for 2023 in areas like cryptocurrency, "buy now pay later," streaming videos, vacation rentals, resale, social media, artificial Intelligence, and cookery in light of the impending recession and persistent inflationary pressures.

2022 offered a fresh set of difficulties for people in the retail, technology, transport, and other sectors as COVID-19 was in the rearview mirror for many companies. Our industry analysts anticipate continued upheaval in significant subsectors, bundling in others, and cost-cutting measures from consumers almost everywhere in 2023.

Despite continued volatility, cryptocurrency will survive

Into 2023, cryptocurrency is arguably the sector with the most open questions. Market collapses in 2022 have put the digital currency on life support; Bitcoin's value has dropped by about a third since the year's commencement. However, despite FTX, one of the biggest cryptocurrency exchanges, going under and damaging its reputation, interest in buying cryptocurrency has remained consistent, and owners continue to believe that the commodity will eventually rebound.

Consumers are rather good at forecasting the health of bitcoin, the bellwether for the larger crypto sector. Therefore, that factor is crucial. Consumers have correctly predicted bitcoin price declines this year, despite overestimating the cryptocurrency's worth. For instance, in January surveys, respondents expected that within six months, the price of Bitcoin would drop

to roughly \$27,000; in July, the price was between \$19,000 and \$23,000. When customers predicted that the price of Bitcoin would drop to \$21,000 again in May, a trend resembling this one started to appear; today, the price is at \$17,000. According to a customer study from November, a price point of \$11,500 is anticipated by May of 2023.

The following macroeconomic reasons contribute to this ongoing decline: A decrepit climate is being created for cryptocurrency to stable, much less develop, because of forty-year high inflation, rising interest rates, and prolonged geopolitical instability. These factors are crucial to the longevity of cryptocurrency.

Despite rising consumer demand, comprehensive regulation is unlikely to happen in the coming year as the Biden administration strikes a balance between monitoring criminal behavior and limiting a potentially lucrative business. The Securities and Exchange Commission is most likely responsible for any regulation.

There will be more proliferation and scrutiny of "buy now, pay later."

While BNPL services have shown consistent growth among Gen Z adults and millennials throughout 2022, this expansion coincided with greater regulatory scrutiny. The offer's attraction is clear when inflation pushes household budgets to the limit: spread out payments for items over a set period, typically with no interest. By collaborating with these suppliers, merchants may offer their customers more flexible payment alternatives and gain sales from customers who can't bear to pay the full price at once.

Following a small post-holiday fall, months of continuous growth in BNPL use among Gen Z adults and millennials were observed after the 2021 winter vacations. Consumers have maintained their spending this year despite inflation, and BNPL loans are especially appealing to shoppers worried about their finances.

During the holiday season, worried shoppers were 20 percentage points more likely to plan to use BNPL than the less anxious ones. These new loan options have demonstrated greater consumer acceptance and are pervasive in retail: Black Friday week saw a 78% rise in BNPL payments compared to the prior week. Importantly, clothes are one of the most popular categories for BNPL use, and these regular purchases will support its growth into the following year.

Retailers and BNPL services should be aware of several drawbacks to this quick expansion. According to a LendingTree survey, consumers need to take out more small loans and make all the payments. Additionally, providers are coming under scrutiny from the Consumer Financial Protection Bureau, and the current interest rate environment is making it difficult for BNPL providers to operate. Retailers should welcome more regulation in this area since it's bad for brand associations for BNPL lenders to be accused of predatory tactics, according to retail and e-commerce analyst Claire Tassin.

Video streaming bundles will re-emerge as a dominant trend

2023 will be the year of the package in the streaming wars. While bundles as a trend seem like a throwback to how new video streaming services used them to gain traction early on in 2019 and 2020, streamers will need to come up with fresh, cost-effective strategies to speed up sign-ups in light of recently revised forecasts for streaming growth. There needs to be more than ad tiers alone to win over wallet share from budget-conscious customers, and this will bring back the good old bundling chances.

While the largest domestic video streamers won't be combined with those from other parent companies in discounted bundles, we'll see more streamers forging bundle agreements with telcos, nonmedia giants, or music streaming services to accelerate growth. Think of deals like the previously offered Disney+ and Amazon Music bundle, T-Netflix Mobile's deal, and the inclusion of Paramount+ in the Walmart+ offering.

The merging of HBO Max and fellow Warner Bros. Discovery-owned streamer Discovery+ in 2023 will be the best example of the strength of two streamers working together. Although the cost of the combined service is presently unknown, it will pressure top video streaming providers to offer more material to customers at a lower cost, according to media and entertainment analyst Kevin Tran.

New competitors threaten Airbnb amid a gloomy housing market.

There are a lot of strong opinions about Airbnb's collapse. Social media is replete with pile-on threads of complaints about the vacation rental company, calling it "Airbnbust," ranging from rage over unexpected prices to worry

about hidden cameras. Despite this, Airbnb has managed to weather several pandemic-related storms and benefited from the new lifestyles and business methods that COVID-19 has brought about.

Travelers may now search "anytime, anyplace" listings, allowing individuals not tied to a specific vacation date or destination to emphasize enjoyment over logistics. Long-term stays create significant income for the company, and all of these elements played a part in the business' record-breaking reservations throughout the summer of 2022.

However, 2023 is expected to be a turning point for the ultimate disruptor of the tourism sector. The difficulties between hosts and guests are familiar to Airbnb, and recent updates have addressed some of the major issues, such as hidden fees and lengthy chore lists. But the distinct position Airbnb holds in the travel industry is threatened by escalating competition from established hotel chains.

If customers still have concerns about costs, duties, and safety, Marriott's relatively new Homes & Villas service offers a comparable experience without the hassles. Consumers no longer perceive Airbnb as a scurrying upstart, which means that tourists will compare items more thoroughly rather than allowing the brand alone to influence their decision. Airbnb has consistently been able to remain ahead of the competition.

A depressing housing market threatens the brand's success even if Airbnb can overcome the competition. The effect of Airbnb (and other holiday rental businesses) on the availability of affordable accommodation has long been the subject of criticism. The issue will receive greater attention if interest rates stay high until 2023, especially among younger buyers, who comprise most of Airbnb's clientele and are priced out of becoming first-time homeowners.

In 2023, Airbnb will have to tackle these problems head-on if it wants to continue to be popular with young tourists. Otherwise, viewers of the potato house advertising would become irritated, according to an analyst of travel and hospitality Lindsey Roeschke.

Growth of the resale market will surge.

Resale has had a successful year this year: brands like Sandro, Timberland, and Athleta have been experimenting with the resale market and are using various tactics, including platform partnerships, peer-to-peer

exchanges, and brand-owned consignment. It remains to be seen which tactics fail and which succeed.

However, the selling tendency is far from reaching its peak in 2022; the framework has been set for a spike in 2023. The significant influx of brands attracts consumer interest because they are ideally positioned to address typical obstacles to consumers' and sellers' participation in the secondhand market, like confirming product quality and authenticity. Connecting customers' purchase histories with resale platforms can help companies make it simpler to sell products they no longer want by reducing the time spent creating listings.

At the same time, buying secondhand goods has become fashionable as buyers find it harder to acquire new ones due to inflation. Women, in particular, reported doing more secondhand shopping in 2022: In November, 22% of women said they had purchased used goods online during the previous three months, up from 18% in May, while 35% of women said they had purchased used goods in-store during that time. Women are more likely to report having serious financial difficulties and are particularly drawn to the cost savings of secondhand shopping.

The good news is that secondhand shoppers are addicted to Christmas goodies, just like this writer. According to a recent Morning Consult analysis, people who have bought secondhand in the past are likely to do so again. As more companies enter this market amid ongoing, elevated inflation, more consumers will be encouraged to try secondhand shopping and are likely to do so again, according to Claire Tassin.

Social media competitors like Mastodon and BeReal will fail.

The competitor social platforms that gained traction in 2022, such as BeReal and Mastodon, will need help to keep up their momentum in 2023, which will irk antitrust enforcers. These two apps, which are very dissimilar, have attracted attention this year for various factors. However, growth will slow in 2023 as social media giants actively promote their BeReal imitation features and more users realize that Mastodon is too complex to be a reliable alternative to Twitter.

These are just two examples, but their likely course has, regrettably, happened much too frequently. It's quite uncommon for a hot new social media challenger to keep getting better after achieving initial success and

establish itself as a serious threat to the market like TikTok did. Just keep in mind "tbh," "Houseparty," or "Clubhouse," maybe the most well-known example.

The emphasis on genuine connections rather than the hyper-curated content that is incentivized on platforms designed around "like" counts is a recurring theme among many of the newest and most successful social media networks that appear. There is something here that will appeal to younger demographics. Therefore social media opponents should continue to investigate this general approach. There is still plenty of room to experiment with how to carry out that vision, though, according to Kevin Tran.

Major tech companies will continue to invest in AI.

Major technology businesses are focusing on investing in cutting-edge machine learning applications that will streamline existing products and services or develop entirely new services shortly, or they already see returns on their investments in these applications. Even though technology hasn't quite reached the point of accurately reproducing human intelligence, this advanced machine learning is commonly referred to as AI in the media. References to these AI applications are being made more frequently, and interest in the subject has been steadily growing (see Google Search Trends).

Why is everything occurring now? The processing capacity that enables these cutting-edge applications has finally entered the mainstream. Chipmakers like Nvidia are marketing specialized deep learning and artificial intelligence (AI) solutions. Intel is making progress toward meeting its deadline for developing powerful quantum computing chips that are expected to lead in a new era of computational power.

Many daily services we use already have AI programs working in the background. Google said that its investments in AI technology were already paying off in its search, translation, and mapping services in the second quarter of this year. Meta said its platforms received content recommendations from AI programs in the second quarter.

Looking ahead, "generative" AI—functions and APIs that can generate content based on user text input—is where some of the most intriguing breakthroughs in this field are taking place. DALLE, Stable Diffusion, and Mid journey are currently well-liked models that make this possible and can produce magnificent landscapes that would win awards. Based on a brief

suggestion, some programs, like GPT-3, may quickly compose full essays, short stories, or opinion pieces.

New and exciting applications of these advanced learning models will emerge in 2023 (and beyond), but only some are looking forward to them, according to a recent Morning Consult survey. Additionally, there will be risks to jobs and complex ethical issues that we are only beginning to scratch the surface of, per tech analyst Jordan Marlatt.

Customers prefer making their food and baking.

In 2022, inflation dominated the news in the food and beverage industry. The largest rates of inflation in the category since the early 1980s were caused by rising food prices, which prompted consumers to use various strategies to reduce their grocery spending, including downsizing to less expensive options and buying less overall out of necessity.

In addition to the fact that food inflation has not yet decreased, consumer mood has generally been declining as we approach 2023. While widespread job losses have not yet occurred, economists and the Federal Reserve agree that they will as we approach 2023.

This economic volatility and insecurity have invaded the kitchen. In 2023, more consumers will attempt homemade baking and cooking projects due to their desire for comfort, independence, and financial savings. Self-sufficiency during the Great Recession took the shape of gardening and even backyard poultry, which were equally appealing regarding sustainability and access to local goods. Sourdough gained popularity across the country during the brief epidemic recession. Social media popularized sourdough baking because it gave people a feeling of community and something to focus their energy and intellect on when they had free time at home.

Consumers won't be dusting off their bannetons next year, but bread-making will probably be replaced with another time-consuming culinary task that may make them feel accomplished and satisfied. In addition to being practical in cost savings, baking and cooking from scratch may also be emotionally reassuring, which will be greatly valued by many in 2023, according to food and beverage analyst Emily Moquin.

2023 Blockchain Trends

Are you curious about the Blockchain startups and trends that will soon affect your business? A Blockchain is an exceptional tool for companies and customers to democratize services and guarantee data security and privacy. Furthermore, Blockchain evolution is spreading across industries due to the increasing demand for cryptocurrencies and Web 3.0 integration. The top nine developments and trends for 2023 are outlined in this section. Tokenization, smart contracts, Blockchain security, and enterprise Blockchain are just a few.

We have examined a sample of 3765 worldwide startups and scale-ups for our in-depth study on the Top Blockchain Trends and Startups. This study provides data-driven innovation intelligence that enhances strategic decision-making by providing an overview of startups and developing innovations in the Blockchain sector. These insights were obtained by utilizing the 2 500 000+ startups and scale-ups globally.

The SaaS platform allows us to identify pertinent startups, new technologies, and future industry trends rapidly and thoroughly because it is the largest data source on emerging companies worldwide. The Top Nine Blockchain Trends will affect the industry in 2023. Startups and scale-ups are developing solutions for accelerating crypto deployment and transactions. Tokenization improves the liquidity of both tangible and intangible assets, while smart contracts enable advanced Blockchain transactions. Blockchain developers are developing novel cryptography and security techniques to enhance data security and dependability. Private and other Blockchain networks designed for enterprise applications are also gaining popularity. Lastly, startups are creating decentralized applications (dApps) for various use cases on Blockchain networks. The US and Europe are the regions with the most startup activity.

In the sections below, you will learn more about eighteen of the three 700+ prospective businesses and scale-ups, as well as the products they create. These Blockchain firms have been carefully chosen based on factors like founding year, location, amount of money raised, and others. Your top choices may look very different depending upon your demands.

Top Nine Blockchain Innovation Trends for 2023

1. Cryptocurrencies

Cryptocurrencies use Blockchain technology to store transactional data in peer-to-peer networks. They do away with centralized institutions like banks to lessen transaction costs and hasten fund transfers. The growing demand for cryptocurrencies encourages the development of high-performance, environmentally friendly coins with shorter transaction times. Additionally, Blockchain adds a built-in security to crypto transactions, enhancing the security of financial transactions. Wallet data is vulnerable to attacks since exchanges typically store it on centralized servers. Cryptocurrency exchanges and patrons will be able to decrease this threat and replace fiat money with cryptocurrencies thanks to advancements in Web 3.0.

Bottlepay offers a Crypto Payment Gateway.

Bottlepay provides a UK-based startup, A crypto payment gateway. Users can convert GBP or EUR to BTC in any direction after onboarding through an expedited verification process. The firm does this by utilizing the Lightning network, a payment mechanism built on Bitcoin. Thus, the gateway enables users to use cross-border transactions without fees to pay content providers through social media platforms.

ChainCard makes a Web 3.0 crypto wallet.

Canadian firm ChainCard creates a Web 3.0 cryptocurrency wallet. It enables users to use cryptocurrencies like BTC and ETH on platforms supporting Blockchain and VISA. The wallet also facilitates asset exchange within the wallet and offers customers a return on their cryptocurrency savings. This makes it easier to replace fiat money with its cryptocurrency equivalents effectively.

2. Asset tokenization

By dividing digital or physical assets into digital tokens, tokenization utilizes Blockchain technology. This is the same concept behind cryptocurrency. However, companies and retail customers can turn assets like real estate holdings and artworks into digital tokens because of Blockchain-powered tokenization.

Due to higher liquidity of previously illiquid assets, owners can ideally sell tokens on secondary marketplaces. This expands investors' access to previously inaccessible assets and offers fresher possibilities for portfolio diversification. Additionally, smart contracts enable one source of authenticity by automating token transactions and elevating transactional transparency. As a result, tokenization makes it available for all parties to check their holdings and guarantee ethical behavior.

Buzzmint simplifies the Tokenization of digital assets.

A firm in the UK named Buzzmint creates a platform for tokenizing digital assets. It makes it possible for businesses and creators to swiftly produce and distribute non-fungible tokens (NFTs) based on their digital assets. The platform additionally enables them to take advantage of current digital infrastructure and stop sending their consumers to other platforms so they may sell their tokens.

Courtyard aids physical asset tokenization.

The US-based startup Courtyard develops a platform for tokenizing tangible assets. Users can vault the underlying assets while creating digital twins of physical assets as NFTs. The software securely preserves the transaction and provenance history on the Blockchain, enabling them to trade NFTs on any market. This eliminates counterfeits while allowing the firms to set up automated royalties and sell goods abroad.

3. Smart contracts

All Blockchain ecosystems rely on smart contracts because they eliminate intermediaries from the governance and execution of all peer-to-peer (P2P) transactions. Startups thus provide simple platforms that let corporations and

Blockchain engineers make smart contracts. They often involve less coding, which quickens the development process. Due to the rise in cyberattacks against Blockchain systems, there is also development interest in solutions that audit smart contracts for weaknesses. Developers of smart contracts can automate security evaluations using such platforms. Smart contracts are more effective thanks to contract structure, access control, and cryptography developments.

Viveel facilitates Smart Contract Creation.

Viveel is a startup from Portugal that develops a smart contract builder for businesses. The platform startup uses visual programming and templates-driven workflow to speed up production. Organizations can implement smart contracts immediately after specifying the components and configurations. The platform audited the deployed smart contracts, making them active decentralized apps. This expedites the time to market and enables the quick deployment of smart contracts in Blockchain networks.

Acria Network builds a cross-chain, real-world data network.

Acria Network, a German firm, creates a cross-chain real-world data network. It uses oracle nodes that employ cross-chain technology to cut out intermediaries and provide access to real-world off-chain data for different Blockchains. The network also enables users to use their choice coins as security. Oracle nodes can only be operated by real-world data distributors, ensuring the reliability of the data. Other participants earn staking rewards for adding to the nodes' reputation.

4. Advanced cryptography

Blockchain networks use cryptography to encipher communication between nodes to ensure the immutability and verifiability of transactions. For this, symmetric and asymmetric cryptographic methods are used by Blockchain developers. Compared to the asymmetric approach, which uses public and private keys for message encryption and decryption, symmetric cryptography provides the same key for the communicating nodes.

Multi-signature algorithms, for instance, produce digital signatures with the help of numerous parties. Another method for proving information without disseminating it across the network is zero-knowledge proof (ZKP). These methods let Blockchain developers maintain security while increasing user and transaction privacy.

Ruby Protocol makes a privacy-centric Web 3.0 Protocol.

Startup in Canada, Ruby Protocol creates a decentralized Web 3.0 protocol that prioritizes privacy. It uses functional encryption (FE), which enables users to encrypt data on-chain that owners of authorized private keys can only decrypt.

The protocol implements a privacy layer that communicates with the multi-chain ecosystem and functions as a private data management framework and data access control gateway. This makes it possible for NFT-gated access, regulatory-compliant asset ownership, private know-your-customer (KYC) authentication, and privacy-preserving cloud computing.

Goldario offers tokens backed by precious metals.

Goldario, an Estonian firm, provides tokens backed by precious metals. The startup tokenizes the eco-cycle of the gold and emerald mining, jewelry production, and retail jewelry industries by combining Blockchain and cryptography. This increases the accessibility and liquidity of valuable commodities and the security of trading in digital assets. Additionally, the startup promotes lower investment thresholds and monitors the development of the businesses they fund to increase transparency.

5. dApps

Blockchain networks with peer-to-peer nodes and smart contracts allow for decentralized applications. The Ethereum Blockchain, for instance, provides specialized tools for developing decentralized apps. dApps provide privacy and development flexibility by removing the oversight and interference of centralized authorities. Additionally, it allows all stakeholders to pay a fair price for applications and services, increasing transparency and

monetization possibilities. Further, as dApps use decentralized computing and open-source licensing to provide a secure development ecosystem, they never go offline. Last but not least, rapid Web 3.0 integration requires decentralized applications backed by Blockchain.

Cube makes dApp token staking easier.

An Indian business called Cube facilitates dApp token staking. The startup's API enables stake access integration for app developers. Additionally, Cube uses asymmetric cryptography and transport layer security (TLS) to encrypt its API calls. It allows for automatic yield tactics like leverage staking and auto-compounding. Developers of dApps can do this to speed up time to market and reduce development time.

Moralis Web 3.0 provides enterprise-grade Web 3.0 APIs.

Swedish firm Moralis Web 3.0 creates enterprise-grade Web 3.0 APIs. The start-up offers APIs for the Solana and EVM Blockchains, as well as for NFTs and streaming Blockchain data. This enables businesses and dApp developers to speed up dApp prototyping and swiftly grow apps. The startup also lessens the expense and difficulty involved in developing dApps.

6. Blockchain-as-a-service

The accessibility and adaptability of cloud-based services support the service business model for Blockchain networks. This decreases time-to-market by enabling Blockchain developers to build and host Blockchain applications and smart contracts quickly. Blockchain, as a service, spares businesses the expense of hiring programmers for internal network development. Thanks to service providers, they can concentrate on improving features and products for their business while maintaining an adaptable and effective network.

NIFTRON produces Blockchain-as-a-service platform.

The UK-based business NIFTRON creates a Blockchain as a service platform. With the help of it and the startup's software development kit (SDK), enterprises may use Blockchain and create plug-and-play modules for their apps. The software also manages user accounts and transactions and enables users to tokenize different data types. Businesses can reduce their development expenses and time thanks to this.

Blockchain solutions are offered as a service by Everblock.

Everblock is a Brazilian firm that offers decentralized storage and smart contracts as a service for the media, education, and healthcare sectors. It enables businesses to use Blockchain-driven applications to protect and increase data transparency. As a result, the business helps media companies combat copyright violations, educational institutions prevent document fraud, and healthcare organizations improve patient privacy and data compliance.

7. Private Blockchains

Permissioned Blockchains, known as private Blockchains, are typically held by companies and organizations. Private Blockchain networks, in contrast to public Blockchain networks, have centralized authorities that control network user access. Private Blockchains also allow for the reversal or deletion of transactions, and each node stores a copy of the complete Blockchain. This gives organizations more control over enterprise data while maintaining the ability to validate and verify transactions. Private Blockchains have fewer nodes than public networks; therefore, they can scale much more quickly and have higher transaction throughput.

AlleoChain promotes the development of private Blockchains.

A Cypriot firm called AlleoChain supports the creation of private Blockchains. Businesses may host private Blockchains in a secure cloud environment utilizing the startup's no-setup, no-code technology, which also uses SQL to manage data integration. The data are also backed up for added redundancy, making it possible for businesses to scale Blockchain applications swiftly.

Dafecs facilitate the deployment of Virtual Organizations.

Turkish startup Dafecs creates a platform for the deployment of virtual organizations. It mixes the InterPlanetary File System (IPFS), tokenization, the internet of things (IoT), a private Blockchain network, and front-end libraries. As a result, the platform secures stakeholder data sharing, enhancing organizational governance and decision-making.

8. Blockchain security

Blockchain mostly has uses in the realms of investments and financial transactions. Additionally, as organizations and users move toward more Blockchain-centric workflows, data integrity and privacy become critical for all Blockchain networks. Startups create Blockchain security solutions to guarantee the security of such networks. They track smart contracts, nodes, and transactions using advanced analytics and Artificial Intelligence (AI) to look for anomalies. This makes it possible to identify rogue users and stop data or asset theft quickly. Businesses that use Blockchain for data management, asset trading, and supply chain management require Blockchain security solutions.

Veridise provides Blockchain security as a service.

A US-based firm called Veridise offers Blockchain security as a service. Employers and Blockchain developers can automate security analysis for smart contracts, Web 3.0 applications, zero-knowledge circuits, and more with the help of the startup's solutions. As a result, they may quickly find weaknesses before using Blockchain technology and mitigate them via internal tool development.

Forta produces decentralized monitoring networks.

A US-based firm, Forta, creates a decentralized monitoring network for securing and managing Blockchain activity. It comprises independent node operators who search through all transactions and changes block by block for

anomalous transactions. As a result, the network can instantly identify dangers and anomalies in governance systems, bridges, NFTs, and other Web 3.0 systems, as well as decentralized finance (DeFi) systems. Then, to avoid financial loss and promptly eliminate dangers, it notifies subscribers of notifications.

9. Enterprise Blockchains

Due to a lack of data control and increased data visibility, businesses are hesitant to embrace public Blockchains like Bitcoin and Ethereum. Even though some businesses rely on public Blockchains, the demand for private and consortium Blockchains is rising. While maintaining other Blockchain advantages like security and traceability, enterprise Blockchain networks give enterprises better control over their data. This enables businesses to integrate Blockchain-based apps while maintaining data compliance. Additionally, Blockchain automatically improves company data security, reducing costly data leaks.

Simplifying enterprise Blockchain development with AstraKode.

A business Blockchain development platform called Network Composer is provided by the Italian firm AstraKode. Businesses can create personalized permissioned Blockchain networks thanks to it. The platform also includes a low-code approach and automated testing to speed up development operations. As a result, businesses can now host their Blockchain networks on the cloud for easier configuration and accessibility.

Zblocks helps construct the dPlat Pro platform for enterprise NFT campaign management, according to the US-based startup. It connects with customer relationship management (CRM) systems and improves NFT lifecycle management. The technology gives user insights based on NFT interactions and supports multichannel NFT distribution for lower bounce rates. Through these capabilities, businesses may improve client acquisition, retention, and reach across all customer lifecycles (ORM).

Blockchain startups, technologies, and trends

Cryptographic and data processing innovations will improve the functionality of smart contracts, which will benefit Blockchain networks as a whole. The energy required for Blockchain transactions will decrease even further when Blockchain-developing enterprises start to lessen their carbon footprint. These remedies will hasten the spread of Blockchain applications across sectors and fuel Web 3.

The Blockchain trends and startups we uncovered during our process of data-driven innovation and startup scouting are only a small portion of the themes included in this study. Digital assets, quantum computing, and enterprise Blockchains, among other things, will change the industry as we currently know it. Gaining a competitive edge entail recognizing new opportunities and integrating developing technologies into your company.

The 2023 Blockchain Startups to Watch

The emerging digital economy is giving Blockchain technology a lot of traction. More than 200 bitcoin exchanges are operating worldwide due to the rising demand for cryptocurrencies.

Many businesspeople have been motivated to use Blockchains by emerging technologies like Web 3.0, 5G, the Internet of Things (IoT), and Industry 4.0. Investors also follow the trend since they believe this area has profit potential.

In recent years, the venture capital sector has invested a sizable amount of money in Blockchain. Examples include Blockchain-based applications for NFT marketplaces, VR and AR metaverses, e-healthcare businesses, and AI.

The market for Blockchain startups is expanding.

According to a Garner assessment, the market will be worth \$176 billion in 2025 and \$3.1 trillion by the beginning of the next decade. These forecasts make it understandable why so many venture capitalists support Blockchain firms.

According to this report, there were 66 Blockchain firms launched in 2012, but that number soared to 777 in 2017. This indicates that the overall number of Blockchain startups is growing annually. Only the first half of 2022 saw 1199 investment rounds, which brought in a whopping \$30.3 billion, more money than venture capitalists raised in 2021.

The next section will outline a few Blockchain-based firms to watch in 2023. We've reviewed an extended range of promising startups, but there are a few to keep an eye on. Let's first look at some intriguing uses for Blockchain technology outside of money transfers.

Blockchain ideas for success

The Blockchain has undergone considerable changes since the launch of the Bitcoin network in 2009, when it was primarily used for money transfers and was a Proof-of-Work token. Today, there are numerous Blockchains with better frameworks that allow utility-based applications to work in a variety of settings. Here are a few applications that tech-savvy businesspeople have already embraced.

Smart contracts

Smart contracts are automatically generated agreements between two parties based on predetermined criteria. When certain criteria are satisfied, the contracts are automatically carried out. At this stage, either side may receive the associated assets.

DeFI

Users can lend or borrow assets using decentralized finance applications without a bank. Through a DeFI application, users can stake their coins and generate interest by trading various financial assets, smart contracts, and cryptocurrencies.

Cybersecurity

IoT devices and Blockchain data encryption techniques can be used together to provide password and authentication information to distant devices, effectively making the network impenetrable.

Healthcare

Individual electronic medical records (EMRs) can be linked to healthcare providers on a Blockchain. This keeps patient data safe and secure while enabling healthcare professionals to prescribe treatments, drugs, and

appointment times.

Supply chain

Blockchain networks can offer total inventory visibility across a network, which makes it very simple to follow the progress of shipments. This facilitates tracking and has the potential to reduce operational expenses by billions.

Governance

Blockchains have been used to build safe electronic voting systems that let voters cast a secret ballot from any device. Electronic voting methods are less expensive than traditional voting systems by reducing the need for paper, and the distance voters must travel.

NFTs

NFTs are digital collectibles that can't be changed into cash. Artists can reach out to the online community on their terms through NFTs. NFT collectors are a way to get into an exclusive club, both online and in the real world.

The 11 best new Blockchain companies for 2023

Fluree: <https://flur.ee>, founded in 2016

What does it do? Fluree is a semantic graph database built on the Blockchain. It lets developers make graphs and concept map with fast search speeds. The company promises that all data can be traced back to its source, and its semantic standards make it stand out from the rest. Users can see their data differently and keep it safe with their main product.

MeetKai: <https://meetkai.com/> founded in 2018

This Los Angeles company wants to make a virtual-reality app with a map of the whole world. MeetKai already keeps track of many real-world places and adds them to its app. The company wants to give people in the metaverse a look at real places. MeetKai uses AI models to recognize places and speech patterns to help with its mission and make an engaging VR experience.

Constellation Network: <https://constellationnetwork.io/>

Founded in 2017

Constellation Network is a Blockchain that can be used by entrepreneurs who want to build a new Blockchain app quickly. It has a high transaction speed thanks to its highly decentralized DAG design and level-zero data transport layer (8-.000 TPS in seven seconds using only six nodes). To further bolster its suitability for scalability in the context of Blockchain applications, the network also provides a method for completing transactions without incurring fees. Because of the features of the Constellation Network, developers will be able to build apps that connect to other Blockchains and help them build their ecosystems.

Metaking Studios: <https://www.metakingstudios.com/> founded in 2017

MetaKing Studios is a AAA game company making Blocklords, a

medieval fantasy grand strategy MMO game where players can own digital assets. The team plans to add DeFi features to the game, giving it an economy that doesn't exist in Blockchain games as yet. The team has partners from all over the world, and their goal is to make Blockchain games that are deep and interesting.

Certik: <https://www.certik.com/> founded in 2018

CertiK is a Blockchain startup whose main focus is security. The company was started by professors from Columbia University and Yale. They use research that has been proven to work to make Blockchains safer, and that's something that businesses that use Blockchain can use. CertiK has implemented AI systems and Formal Verification to deal with cybersecurity threats. It also checks the security of other projects and lists the most secure websites on its website.

Poko: <https://www.pokoapp.xyz/> founded in 2022

Poko is an out-of-the-box idea for making DAOs legal. The people who started it call it "Slack for DAOs." It makes it easier for anyone to join an existing DAO or start their own by simplifying the legal and technical steps. The legal setup fee will be reduced to a small amount, like \$50. This will make DAOs available to a lot more people online.

Soon: <https://www.soon.app/> founded in 2020

Soon is different from other cryptocurrency investment apps since it does not require users to be active investors. Soon, anyone with a debit card or savings account can invest without worrying about how their money is being managed. The catch is that the app has a transaction fee of over 3%, which is high. But if you want to invest quickly and easily from your bank account, including investing in cryptocurrencies, a tiny price to pay. The app will start in the United States and then move to other countries.

Payourse: <https://www.payourse.com/> founded in 2019

Payourse, based in Africa, has the best idea out of all the startups there: making tools to set up crypto exchanges. They have set up a website called Coinprodile that lets anyone send cryptocurrency to people in Africa. But Simpa is the thing that people are talking about. It is an app that lets tech entrepreneurs set up crypto exchanges quickly without writing any code. The African crypto market needs to grow the use of digital currencies on the continent.

Finnt: <https://www.finnt.com/> founded in 2022

Finnt wants to make an app that makes it easy to save money and send money back home. Users can set up high-interest savings accounts through a decentralized network, and Stablecoins will be used to back these accounts, protecting those who have them from inflation. Finnt has the potential to become the money transfer app of choice in many countries because it can be used to save and pay across borders.

Argo: <https://watchargo.com/> founded in 2020

Argo stands for "Artists Going places". Artists and people who make short films can share their work on Argo, which has NFT functionality. In fact, it has curated lists for genre-based and niche-based content. It also helps creators get the word out about their movies and make money through NFTs.

JustCarbon: <https://www.justcarbon.com/> founded in 2021

JustCarbon is a market for projects that remove carbon, making the carbon market clear. On the JustCarbon Blockchain, buyers can buy JCR tokens to support projects that are good for the environment. One JCR is the same as taking one ton of carbon out of the air on earth. Even though the company admits that carbon emissions are often unavoidable, they help companies find ways to reduce their carbon footprint.

Many more Blockchain-based startups are trying to use Web 3.0 technology to solve many problems. Even though we've only talked about a small number of these startups, we hope we've provided an idea of where the

Blockchain tech industry is going.

Watch For Blockchain Trends In 2023

Blockchain alters how transactions and data are shared in many industries, including information technology, finance, government, and healthcare. Blockchain supporters have been excited for a long time about the possibilities of distributed ledger technology for cybersecurity, financial services, the internet of things (IoT), real estate, and managing the supply chain.

Looking ahead in 2023, there are a few additional markets to keep an eye on to see how Blockchain is going to be used. Here are four industries that will speed up the use of Blockchain to their great benefit.

Cloud services

International Data Corporation says that the worldwide public cloud services market, which includes infrastructure-as-a-service, platform-as-a-service, and software-as-a-service, made \$408.6bn in 2021. Even though the cloud services industry is still young, it is growing rapidly, which drives new ideas.

For example, "cloud storage" is another area that Blockchain can change and make more electric. Usually, cloud storage companies let users buy a certain amount of storage, which is usually sold by the terabyte as an article on InvestmentBank.com explains. It also said that "Blockchain technology gives users a way to store information on a peer-to-peer network." The nodes that help make transactions possible could run a cloud storage platform, and these nodes would let the consumers use the storage space on their computers. However, customers would pay the nodes' owners for storing their data.

Generally, Blockchain makes privacy and security better for cloud services as a whole. When Blockchain technology is added to cloud computing, the main problems of security and privacy are solved. In a Blockchain network, when you delete data from one computer, it doesn't delete the data from other devices.

So, there is no chance that data will be lost or changed, and data on a Blockchain can't be taken off. With better privacy, security, and automation,

cloud computing can move quickly forward, giving users more benefits and possibly saving providers money on costs and overhead.

Governments

Governments will likely start putting systems based on distributed ledger technology (DLT), replacing paper-based systems. The switch to digital data systems has been going on for a while. Still, DLT has more benefits, such as encryption and validation features that make trust, transparency, and security better.

With the innovative use of Blockchains, government organizations can quickly improve how they communicate with and help their constituents. For example, Blockchain can make it easier to:

- Process government forms and track their chain of custody
- Find and fix government document mistakes, prevent tampering, and find fakes.
- Better operation by getting rid of paper records.
- Controlling costs by eliminating the need to process applications, forms, and other paperwork by hand and using digital records stored on the Blockchain instead.

Law

As Blockchain changes how government works, it will also change how law and legal services are provided. There are already a lot of ways in which the technology is changing the way legal services are done. Smart contracts, settling disagreements, protecting intellectual property rights, etc. are all areas where the Blockchain can be used to prove creation, first use, rights management, and track distribution. This will keep people from stealing other people's work and help plaintiffs in court if copies of the work are time-stamped.

Meanwhile, contract lifecycle management solutions help attorneys deal with how complicated and changing contracts are. This makes it easier for them to complete, sign, and uphold contracts. Blockchain will help law firms

and legal departments in 2023 and beyond as more transactions and records shift to digital workflows, digital signing ceremonies, and digital storage.

Identity

Last but not least, it becomes increasingly crucial to manage and validate our digital identities, which correspond to who we are in the real world, as we progress toward digital commerce and communication. There are a lot of problems with the way identity systems work right now. They need to be fixed, work alone, and make mistakes. Blockchain systems can solve these problems and make it easy to check someone's identity and assets from a single place.

Blockchain identity can also give people a new kind of "self-sovereignty." Web 3.0 is all about giving people control over their personal information and digital data across platforms. Web 3.0 starts with Blockchain, but its goals and goals will cover a large part of the internet, metaverses, and shared networks.

Blockchain is innovating how we do business, share information, and identify ourselves in many fields, including cloud services, cybersecurity, IT, government, healthcare, legal services, real estate, supply chains, and more. These trends are just the beginning. In the coming years, Blockchain will have more chances to make our digital data more private and safer.

Key Features in this Chapter

- Researchers examine key trends for 2023 in areas like cryptocurrency, "buy now pay later," streaming video, vacation rentals, resale, social media, artificial intelligence, and cookery in light of the impending recession and persistent inflationary pressures.
- However, growth will slow in 2023 as social media giants start to promote their BeReal imitation features more actively and more users realize that Mastodon is too complex to be a reliable alternative to Twitter.
- New and exciting applications of these advanced learning models will emerge in 2023 (and beyond), but only some are looking forward to them, according to a recent Morning Consult survey.
- There are Blockchain trends in 2023 in many industries, including information technology, finance, government, healthcare, and more. Blockchain is altering the way transactions and data are shared.
- Governments will likely start installing distributed ledger technology (DLT) systems, replacing paper-based systems.
- Law as Blockchain changes how government works; it will also change how law and legal services are provided.
- Contract lifecycle management solutions help attorneys deal with how complicated and changing contracts are.
- There are many problems with how identity systems work right now.

THE BLOCKCHAIN INDUSTRY HAS A BRIGHT FUTURE

Winter will end and spring will come; and just like it's always darkest before dawn, the current crypto winter that's giving the global Blockchain industry a bad start this year will be over. The crypto sector has stumbled and will do so again as it develops as a new, disruptive technology. Each journey should not be viewed as a failure but rather as a chance for improvement and reinvention.

History is replete with examples demonstrating how humanity's unwavering desire for improvement constantly overcomes hurdles. The great philosopher Nietzsche famously stated that "mistake has made man so deep, sensitive, and innovative that he has thrown out such flowers as religions and arts" in his meditation on human nature.

I can identify with this remark because it is an experimental error that propels us onward and higher. Although societal advancements can never be perfect, we work to achieve this ideal because we will never accept mediocrity.

History has often shown that humanity can light a torch on the darkest nights by relying on the light of creativity. This was true for crucial eras, including the Renaissance, Industrial Revolution, and the 1990s internet boom. I will therefore continue to have faith in the Blockchain industry's bright future. Among many other cutting-edge capabilities, Blockchain technology offers 1) trustless transaction verification, 2) immutability and transparency, and 3) immediate P2P settlement.

Blockchain has recently seen a surge in acceptance among technologists, financial institutions, and mainstream groups. According to data from

CoinMarketCap, the number of cryptocurrencies is roughly four every day, and the NFT market has grown more than 20 times since 2020. Politicians, businesspeople, programmers, gamers, and artists from all walks of life come together around Blockchain because they all share confidence in it and its potential to create a new way of living in the future.

But the potential of the Blockchain sector goes far beyond its scientific advances. Future societal operations could be substantially altered by the revolutionary notion of Web 3.0, for instance. Its potential to introduce the decentralized autonomous organization (DAO) as a brand-new type of international organization is equally attractive. As an illustration, in less than a year after its launch, TRONDAO, one of the busiest Blockchain platforms, has grown to become the biggest worldwide crypto DAO.

As of August 2022, it had more than 108 million registered user accounts and 1,400 decentralized applications deployed on the Blockchain. TRONDAO's flourishing ecosystem is nothing but a witness to the unifying strength of this shared objective. TRONDAO's mission is to accelerate the decentralization of the internet through the use of cutting-edge Blockchain technology. I'm proud of the global individuals who contribute to the TRONDAO ecosystem as users, developers, and supporters.

The Blockchain sector has a promising future as long as people keep innovating and pushing one another to new heights. On the decentralized web, we will never cease fostering new connections. Together, we won't ever stop building. And we'll never cease building a more solid basis for a better future.

Blockchain on the Verge of a Revolution in Investment

The term "Blockchain" was originally used in a written report published in 2008 by "Satoshi Nakamoto," the alleged pseudonymous individual or individuals who developed Bitcoin. At the time, "Bitcoin" was the name of the proposed peer-to-peer electronic cash system. With over 100 million users worldwide, Bitcoin has emerged as Blockchain technology's most popular application. However, because the underlying technology is rapidly proving to be a disruptive breakthrough, Blockchains can have many use cases beyond digital currency. A Blockchain is a distributed, decentralized public database that is cryptographically secure and maintains an immutable record of all previous transactions.

Database composition: traditional database vs. a distributed ledger

Technically speaking, a Blockchain is a type of distributed ledger technology (DLT) in which data is logged and kept in immovable units called blocks. The ledger is updated, and new blocks are added once a current block is filled. These blocks are safely connected (or chained) using cryptographic hashing. A powerful database is produced with a complete, verifiable, and immutable history of all previous transactions because each new chronological block of data references the previous one. Across an extensive network of computer systems or nodes, this database of transactions is freely transmitted and kept.

The databases throughout the entire network are likewise updated as blocks are added, and the ledger is updated, ensuring that every node has the same copy of the database.

Cryptographic proofs guarantee the data's accuracy and do away with the requirement for a centralized, trusted party like a bank. It is effectively "decentralized" because the data is spread across numerous nodes, and no single authority has control over or access to the database. There isn't a single point of failure as a result.

Blockchains make it possible to securely transfer digital assets without needing a centralized party to mediate the transaction. Although there are many different Blockchains and the technology is developing quickly.

Generally, a Blockchain-based database is digital, decentralized, uses cryptography, has a consensus mechanism, and is immutable, real-time, and secure.

Association with cryptocurrency

Blockchain technology is being used for the first time in cryptocurrencies like Bitcoin. But there are now more applications for decentralized networks than just digital money. Not every Blockchain is a cryptocurrency, even though all cryptocurrencies are based on Blockchain technology. Because of its core characteristics, such as enabling untrusted parties to reach a consensus and enabling the safe proof of ownership of digital assets through encryption and digital signatures, Blockchain technology is the foundation for the existence of cryptocurrencies.

The inability to change or remove transactions from the Blockchain contributes to the rarity of digitally native assets and avoids double-spending. Cryptocurrencies would not be possible without Blockchain technology.

Bitcoin has drawbacks regarding transaction speed, fees, and volatility of its underlying value, despite being utilized by some as a sort of "peer-to-peer electronic cash" or medium of exchange. Due to its small block size and poor processing speed compared to high throughput payment networks like Visa or Mastercard, Bitcoin can only handle 7–10 transactions per second. Furthermore, the price of a Bitcoin transaction can frequently be more than a few dollars, depending on the demand for block space.

For big-value transactions, this is okay, but the costs are too costly for frequent, smaller-ticket purchases like those made through a credit card network. As a result, it is unlikely to be extensively used as a form of payment to exchange goods and services from various merchants.

We think that bitcoin's strongest application and the best product-market fit is a safe store of value. This idea is supported mainly by bitcoin's rarity, longevity, and adoption history. A total of 21,000,000 Bitcoins will be "mined" or created, and the Bitcoin network's encryption has been thoroughly tested over 13 years and found to be both secure and unbreakable. According to the "Lindy Effect" hypothesis, something's expected to last longer the longer it has been around. According to this standard, Bitcoin has established itself over a considerable amount of time, particularly by modern technology standards. It is likely to be increasingly recognized as an investible asset (like

gold) and develop into a workable asset to store value for all investors worldwide.

Blockchain enables “smart contracts”

Ethereum and its native token, Ether, are the second most well-known uses of Blockchain technology. Ethereum is recognized as a second-generation application of the technology established with customization and programmability as a first premise if Bitcoin is seen as the first generation of Blockchain technology.

The broad programmability of Ethereum is far superior to Bitcoin's first-generation capability of straightforward balance transfers.

The idea of "smart contracts," or small computer programs that are distributed and run on a Blockchain network, was invented by Ethereum. These smart contracts, known as DApps (Decentralized Applications), make it possible for Ethereum to function as a "global internet computer" that never sleeps and can run various applications like your smartphone or home computer. While alternative Blockchain networks that support smart contracts and DApps have subsequently emerged, Ethereum continues to be the most widely used one, with over 80% of DApp activity taking place on top of the Ethereum network.

The greatest ecosystem of developers creating tools and programs is likewise focused on Ethereum. ERC-20 smart contracts, which take advantage of the security, data accessibility, and execution capabilities of the underlying Ethereum Blockchain, are present in many of the most frequently traded tokens.

Smart contracts are increasingly facilitating all kinds of financial transactions. Also known as DeFi, this group of on-chain applications and protocols offer financial services and are constructed on top of decentralized networks without centralized intermediaries (decentralized finance).

DeFi will replace some roles of the traditional financial intermediaries.

The field of traditional financial services, or "TradFi," is home to one of the most promising and potentially revolutionary Blockchain applications. The ecosystem of decentralized finance, also known as "DeFi," can

completely rewire our financial system and give it significantly higher levels of efficiency, openness, justice, and inclusivity when compared to "TradFi." The current banking sector accounts for around 8% of the world's GDP, so DeFi is a considerable possibility.

Decentralization occurs in various financial services, including lending, borrowing, exchange trading, derivatives, yield farming, staking, insurance, and asset management. The goal of DeFi is to make financial services more widely available and to reduce the influence of centralized institutions. With all the advantages of a public Blockchain, which directly addresses many of Traditional Finance's major drawbacks, DeFi also aims to enhance "TradFi." DeFi stands in stark contrast to TradFi because it is decentralized, trustless (i.e., doesn't demand blind trust in a third party), permissionless, composable, and transparent. TradFi is centralized, requiring trust in third-party intermediaries, has disparate systems and regulatory hurdles, and is opaque.

The Top Blockchain Implementation Challenges for 2023

Investors find Blockchains attractive because they represent cutting-edge technology with the potential to reduce transaction costs significantly. Blockchains enable safe, direct exchanges between an unknowable number of individuals, who may or may not be trustworthy of one another.

It can be easy to forget that Blockchain technology is still in its infancy, given how far it has come since its creation. Ten years can seem like an eternity in the Blockchain sector, yet early-stage projects are often characterized by rapid growth. As a result, the technology still has a ways to go before realizing its full potential. It must also become widely accepted, especially in professional contexts.

Although the Blockchain craze phase has gone, other implementation-related issues are starting to appear. The difficulties in implementing Blockchain technology in 2023 will be covered next.

Challenges of implementing Blockchain in 2023

1. Increasing cost of implementation

It all depends upon how much money is invested upfront. Some businesses need help to afford the cost. Even though most current solutions are free, there are still significant costs associated with hiring qualified software developers who are experts in Blockchain construction, licensing fees when switching to a billable software version, general administration, and more. It is a significant barrier to the implementation of Blockchain technology.

2. Failure in adoption

When many individuals collaborate, Blockchains are most effective and valuable. A Blockchain ecosystem requires customers and providers to join the network for it to operate, and Blockchains will only be practical or scalable if many people use them.

3. Problems with privacy and security

Despite the anonymity afforded by cryptocurrencies, many potential Blockchain applications need smart transactions and contracts to be undeniably related to real people, which presents severe privacy and data security concerns. Here, Blockchain technology might be applied, either on a private network or one managed by a consortium. Access would be restricted to what is required to maintain your private data's privacy.

4. Restrictions on HR capacity

People that can conceptualize, apply, and integrate Blockchain technology are needed in business. Everyone is included in this, from the CEO to the product manager and up to certified developers. Few people in India are proficient in both technology and business. Blockchains can be created, implemented, launched, and write smart contracts, but there is a serious shortage of certified Blockchain developers or engineers.

5. Criminal activity

Due to the absence of laws and the fact that Blockchain technology is still in its infancy, several shady companies have sprung up to take advantage of the trusting public. The examples of bitcoin theft in recent years have drawn a lot of attention and almost brought the entire cryptocurrency industry to an end.

6. Increased energy usage

It takes plenty of effort to put Blockchain technology into practice. This restricts access to PoW networks for regular users and encourages the creation of enormous mining pools, which undermines decentralization by obliging members to join huge mining pools.

7. Interoperability

Interoperability is one of the most urgent issues that must be solved before businesses completely embrace Blockchain technology due to worries about compatibility between various Blockchains. Most are separated from one another and do not communicate with other peer networks because data can't be sent to or received from them.

8. Deficiency in standardization

Despite the wide variety of networks available, there is no global standard. We are seeing problems with incompatibility, growing costs, and complicated procedures because there is no worldwide standard. The lack of a standardized Blockchain implementation deters potential contributors and backers.

9. Data validation

A distributed, digital ledger that cannot be changed once created is called a Blockchain. This information must be secure, ideally from a single, stable source, preferably automatically, without human intervention. If the data is input by a human prone to error, more work will need to be done to validate the data. Because of this, great care must be taken to ensure that the data being added to the Blockchain is correct and trustworthy.

10. Awareness

We've established that both the Blockchain and the DLT ecosystem are relatively new and have great potential. Given that news about price changes in the cryptocurrency market frequently makes the front pages of traditional media outlets, it is not surprising that the general public is unaware of the immense benefits of the technology that underpins cryptocurrencies.

Blockchain Technology Could Revolutionize Healthcare in the Future

The way we contemplate the future of information and data has been completely transformed by Blockchain technology. At its most basic level, it provides for the decentralization of data or the transfer of information control from a centralized entity (such as an individual or organization) to a dispersed network. Using a distributed network for information control allows society to improve the data's transparency, trust, and integrity.

According to the authors of a white paper published by the Sutardja Center for Entrepreneurship & Technology at the University of California, Berkeley, Blockchain is essential because it is a distributed database of records, or public ledger of all executed transactions or digital events to be shared among participating parties. The consensus of a majority of the system's users verifies each transaction on the public ledger. Additionally, data that has been input cannot be deleted. The renowned cryptocurrency Bitcoin, a decentralized digital currency, has been the most widely used Blockchain application.

Blockchain advocates contend that the technology will transform all businesses by facilitating safer, more reliable, and easier data and information sharing. While some industries are eager to be among the first to adopt new technology, others are more cautious about how it should be applied and are moving slowly.

In a presentation given at the end of 2017, the US Department of Health and Human Services (HHS) explored the potential applications of Blockchain technology in the healthcare system. The study provided numerous possible applications for its use, including supply chain transparency, securing access to medical records, expediting insurance company communications, and even better-enabling capabilities for remote patient monitoring.

Blockchain proponents think the technology can offer greater security and authenticity than current alternatives, particularly in healthcare data. "Blockchain-based solutions for health documentation offer secure encryption mechanisms that guarantee the confidentiality of patients' information when connecting with other parties," according to an article from the World Economic Forum.

Pre-authorization will be drastically reduced by tokenization, smart contracts, and the encryption methods used in Blockchain network

transactions, allowing patients to receive the necessary and informed care more quickly. When previously relying on the patient or files physically delivered or emailed from many sources, such as local physicians, labs, etc., the healthcare professional can get pertinent information promptly.

Others are cautious: one reason is that Blockchain needs to be better understood. This makes people skeptical of and reluctant to use the technology for something as important as the healthcare infrastructure. Furthermore, despite the expansion, Blockchain technology has few use cases. The technology itself is not new, but its foundation is. The HHS presentation clarified that modern technology has only begun to take shape after variants of the original Blockchain hypothesis were first introduced in the 1980s.

Innovation in healthcare information and data is long needed, especially in a world where cybersecurity risks and data fidelity are perennial problems. Blockchain technology is undoubtedly here to stay, but innovators, decision-makers, and tech enthusiasts must develop methods to integrate it into healthcare in a secure, accountable, and patient-focused way.

Security Level That Users Deserve

A recent DARPA report highlighting how bitcoin technology is susceptible to hacking makes it clear that while Blockchain technology was established partly on the premise of high-level security, this is not necessarily the case. Given this context, Blockchain service provider that go far beyond the worlds of cryptocurrencies and NFTs should provide clients with the level of security they are entitled to and, in some cases, believe they already have.

How Blockchain security achieved its current state

Cryptocurrency is proven to be a gold mine for hackers. Attacking Blockchain "bridges," devices used to move tokens across Blockchains, is the most recent fad among hackers. In the first seven months of 2022, these Blockchain bridges have seen thefts totaling more than \$1 billion.

Breaches are frequently the result of poor key management. People enjoy some freedom when given power, which Blockchain appears to do, but it also carries some responsibility. For instance, even if an organization has strong user-level security controls, hostile actors can still deceive individuals into signing documents or handing over private keys.

It's not so much that Blockchain as a technology has flaws at its core; instead, it has produced a situation in which people who are not security experts find themselves in charge of valuable items and end up as targets of skilled hackers.

Instead of creating additional protocols to solve the gap, more user-level controls should be implemented. There could be five levels of approvals, albeit the number will depend on the size of the business. A transaction of a given value might need the permission of several persons in several places. Dapps on Ethereum give the platform some programmability, but they need clarification because users are unsure of whose apps they are signing or allowing.

We Must Invest in a Green Blockchain if We Want it

The need for sustainability of Blockchain technology is one of its many detractors. Here are some potential improvements. Blockchain storage is quickly becoming a legitimate rival to the cloud. The distributed nature of Blockchain storage makes it a more secure type of storage than keeping your data on the cloud, which is one unquestionable benefit it offers.

However, high energy consumption and a considerable carbon footprint have consistently hampered Blockchain technology. Environmentalists have a particular beef with cryptocurrencies. For instance, it is well known that Bitcoin uses more electricity annually than whole countries. The future of energy-intensive Blockchain technology may become irrevocably tied to its capacity to offset its carbon footprint at a time when a United Nations body recently declared that the last eight years had been the warmest ever documented in modern times.

What's holding back Blockchain companies' environmental, social, and governance (ESG) transformation?

Companies, including the formerly ESG-resistant FAANG, have now swung around to commit to concrete targets surrounding their ESG aims as discussions about responsible consumption are raging. Even Morgan Stanley stated that to determine a company's development potential, ESG-focused measures may rule the next ten years of investment.

But even while ESG metrics influence investment decisions, it is essential to remember that some people may find it more challenging to make the right ethical decision. While some of the biggest international corporations, like Apple and Google, can afford to switch their attention to environmental, social, and governance issues with relative ease, the same is only sometimes true for businesses with a Blockchain focus, even the more well-known players.

Institutional investors are inconveniently out of reach for most cryptocurrency projects since they are under more intense scrutiny than ever for ESG reporting. This, in turn, impacts the overall momentum of the broad adoption of mainstream Blockchain technology. Businesses operating in a

dispersed environment with tens or hundreds of Blockchain-related servers lack the latency necessary to commit to ESG.

The Blockchain industry needs to focus on a wider audience.

Blockchain, especially bitcoin, has formed a core group that is made up of 94 percent of younger millennials and GenZ due to its anti-establishment taste. But the technology must appeal to a broader audience to see widespread adoption and investment.

It is commonly known that younger investors are more likely to choose riskier investments, such as in cryptocurrencies, which are renowned for their rapid price changes. Many middle-aged and older consumers have little interest in this type of risk because it does not appeal to people trying to save for a home, family, or retirement.

Even many members of Gen Z and millennials, the generations thought to be the most climate-conscious to date, decide against using Blockchain technology because of the potential harm it can cause to the environment.

Due to the tiny audience, it is difficult for big businesses or those aiming to make big profits to invest in the technology. This prevents the development of greener initiatives because many companies in the sector may only be trying to survive.

Blockchain technology must demonstrate applications outside of cryptocurrencies. As Blockchain storage steadily acquires commercial acceptance as a more secure alternative to the cloud, this picture makeover will likely take place over time.

A greener Blockchain is possible

The Blockchain sector is currently poised on the cusp of widespread acceptance. Each month, it may easily add thousands of new users. But for Blockchain to go beyond an emerging technology and into the mainstream, organizations need capital to secure ESG initiatives and fascinate the largest audience.

Projects like the Green Treasury Initiative by ClimateTrade, which increases the number of Blockchain use cases with a net negative carbon footprint, are examples of how solutions to create an equitable and

sustainable future for Blockchain technology are already beginning to emerge.

Ethereum has plans to update its energy-guzzling machinery, which could reduce its energy use by 99.95%. However, for the sector to achieve widespread acceptance, offsetting the carbon effect of Blockchain networks is likely to be a challenge.

Smaller Blockchain businesses will need enough cash to develop suitable solutions that keep them ecologically friendly. We need to support Blockchain and Blockchain firms so they can have the funds to come up with these solutions if we want to use the Blockchain for good without harming the environment. You must concentrate on ESG activities or collaborate with established cloud companies that are leaders in ESG if you want to reach a much wider audience.

Key Features in this Chapter

- According to data from CoinMarketCap, the number of cryptocurrencies is roughly four every day, and the NFT market has grown more than twenty times since 2020.
- Less than a year after its launch, TRONDAO, one of the busiest Blockchain platforms, has grown to become the biggest worldwide crypto DAO.
- Given this context, Blockchain service provider that go far beyond the worlds of cryptocurrencies and NFTs should provide their clients with the level of security they are entitled to and, in some cases, believe they already have.

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CONCLUSION

This book has provided a thorough and insightful exploration of the world of Blockchain technology and its numerous applications. From its origins as the foundation for the cryptocurrency market to its potential to revolutionize industries such as finance, supply chain management, and voting systems, Blockchain has proven to be a game-changing technology with vast potential.

As we have seen throughout the book, the adoption of Blockchain technology is still in its early stages, and there is much room for innovation and growth. However, Blockchain has the potential to fundamentally change the way we operate business and interact with one another.

The possibilities of such technology to revolutionize the way we do business and communicate are both exciting and daunting. You now have the tools to understand and leverage the power of this innovative technology. From smart contracts to distributed ledgers, you've been given a comprehensive look at the technology and its implications for the future. With the knowledge you've gained, you can now develop strategies to help you capitalize on the opportunities presented by the Blockchain revolution.

"Blockchain" is a system for keeping public transactional records across numerous databases in a network connected by peer-to-peer nodes. The information in the digital ledger is very secure and can be compared to a Google spreadsheet. It uses a digital signature function to execute fraud-free transactions, making it easier for other users with a specific digital signature to corrupt or edit an individual's data.

Blockchain is changing how contracts are enforced and making the

government run more effectively. The next logical explosion in Blockchain applications is the Internet of Things (IoT). IoT offers a wide range of uses and several security issues.

IReporters has compiled 34 real-world application cases for this practical yet ground-breaking technology. The collapse of FTX has sparked worries about the long-term sustainability of digital assets and the industry as a whole. Financial advisors are split on the sector's future, with some advising clients to avoid investing in cryptocurrencies today.

Some more popular currencies, such as bitcoin and Ether, appear resilient, but there is unquestionably a significant shrink in the crypto industry. Experts predict an excellent year in 2023, followed by an even better year in 2021 and 2022.

Investing in the industry carries significant risks for investors. Use a crypto wallet instead of leaving your funds on the crypto market. Amazon, IBM, Intel, and AMD are well-known companies that use or collaborate with Blockchain technology. Coinbase Global, Inc. (NASDAQ:COIN) is a fantastic investment opportunity for investors who want to learn more about the cryptocurrency and Blockchain industries. As of November 30, the corporation reported having \$61.7 million in cash.

The total value of these positions increased from \$13.63 million in the last quarter, with 13 positions to \$29.36 million this quarter. In Web 2.0, you don't have to be a developer to participate in the development process. 2023 will be the year of the package in the streaming wars, with discounts and video streaming bundles. Airbnb's position in the travel industry is threatened by escalating competition from hotel chains. Major tech companies will continue to invest in cutting-edge machine learning applications.

Tokenization allows businesses to produce and distribute non-fungible tokens (NFTs) based on their digital assets. Smart contracts enable a single source of truth by automating token transactions. Private and consortium Blockchains give enterprises better control of their data. The market for Blockchains is expected to be worth \$176 billion by 2025 and \$3.1 trillion by 2023. In short, Blockchain is changing the way transactions and data are shared. It can fix document mistakes, prevent tampering, and find fakes. In the coming years, we will have more chances to make our digital data more private and safer. Here are four industries that will benefit from distributed ledger technology.

As we move forward, it will be necessary for individuals and

organizations to stay informed about the latest developments in Blockchain and consider how it might be used to improve their operations. Whether you are a business owner, an investor, or simply someone interested in technology and its potential to shape the future, "New Blockchain" has offered a wealth of information and inspiration. We hope that it has sparked your curiosity and helped you better understand this exciting new technology's possibilities.